

The Role of Interfaces in Polyethylene/Metal-Oxide Nanocomposites for Ultrahigh-Voltage Insulating Materials

Amir Masoud Pourrahimi,* Richard T. Olsson, and Mikael S. Hedenqvist*

Recent progress in the development of polyethylene/metal-oxide nanocomposites for extruded high-voltage direct-current (HVDC) cables with ultrahigh electric insulation properties is presented. This is a promising technology with the potential of raising the upper voltage limit in today's underground/submarine cables, based on pristine polyethylene, to levels where the loss of energy during electric power transmission becomes low enough to ensure intercontinental electric power transmission. The development of HVDC insulating materials together with the impact of the interface between the particles and the polymer on the nanocomposites electric properties are shown. Important parameters from the atomic to the microlevel, such as interfacial chemistry, interfacial area, and degree of particle dispersion/aggregation, are discussed. This work is placed in perspective with important work by others, and suggested mechanisms for improved insulation using nanoparticles, such as increased charge trap density, adsorption of impurities/ions, and induced particle dipole moments are considered. The effects of the nanoparticles and of their interfacial structures on the mechanical properties and the implications of cavitation on the electric properties are also discussed. Although the main interest in improving the properties of insulating polymers has been on the use of nanoparticles, leading to nanodielectrics, it is pointed out here that larger microscopic hierarchical metal-oxide particles with high surface porosity also impart good insulation properties. The impact of the type of particle and its inherent properties (purity and conductivity) on the nanocomposite dielectric and insulating properties are also discussed based on data obtained by a newly developed technique to directly observe the charge distribution on a nanometer scale in the nanocomposite.

1. Introduction

Nowadays, most electric power is produced through burning fossil fuels that result in a large amount of greenhouse gases (mainly CO₂) detrimental to global warming and climate change.^[1] Substitution of fossil fuels with environmentally

friendly and renewable energy sources is therefore a promising way of reducing the CO₂ emissions. A major challenge is that the solar, wind, and hydro energy sources are often located far from densely populated areas, which means that electricity generation from these distant renewable energy sources must be combined with cost-effective power transmission systems.^[2] High-voltage direct-current (HVDC) transmission shows an energy loss $\approx 50\%$ less than conventional high-voltage alternating-current (HVAC) overhead lines at comparable voltage and transmission distances.^[3] The HVDC supergrids, utilizing extruded cable systems with polymeric insulation, show promising capacities to expand across the sea and underground, where overhead transmission networks are impossible or undesirable.^[4] Currently, the most powerful extruded HVDC cable with polyethylene as insulating material is able to withstand an operating voltage of up to 640 kV with a transmission capacity of 3.1 GW.^[5] To further reduce the transmission losses so that intercontinental cable lines can be used, e.g., from the Middle East and North Africa to Europe, the voltage levels have to increase.^[6] It is anticipated that a HVDC cable system with an operating voltage of 1 MV and a capacity to transmit 5 GW will be available on the

market by 2030, without any change in today's high voltage system's grid geometry,^[7] but these higher voltage levels inevitably impose a considerable electrical stress and increased temperature that may result in breakdown in current HVDC insulating materials based on pristine polyethylene. The heat generation (ΔT) in a cable insulation is proportional to the DC conductivity (σ) and the square of the applied voltage (U), hence an insulating material with ultralow DC-conductivity that prevents the risk of thermal runaway and electrical failure at higher voltage-levels, is required.^[7,8]

The electrical conductivity in an insulation material depends on the amount/concentration of charge carriers (N_i), the charge of the charge carriers (q_i), and their mobility (μ_i), which can be expressed as:

$$\sigma = \sum_i N_i q_i \mu_i \quad (1)$$

Dr. A. M. Pourrahimi, Dr. R. T. Olsson, Prof. M. S. Hedenqvist
KTH Royal Institute of Technology
School of Chemical Science and Engineering
Fiber and Polymer Technology
SE-100 44 Stockholm, Sweden
E-mail: ampo@kth.se; mikaelhe@kth.se

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The charge carriers referred to here are species that can move and carry an electric charge; e.g., holes, electrons, ions, and polar molecules. In order to reduce the conductivity, it is equally important to decrease the concentration of potential charge carriers by increasing the chemical cleanliness of the insulating material, and reduce their mobility by, e.g., the introduction of a filler that contain traps for the charge carriers.^[8] Recently, polyethylene nanodielectrics (nanocomposites) based on an ultralow loading volume of metal-oxide nanoparticles (<1 vol%) have been introduced as a potential material group for electrical insulation materials due to their lower charge mobility and DC conductivity than pristine polyethylene.^[9–13] The concept of “nanodielectrics” was first introduced by Lewis in 1994^[14] and refers to composite materials containing an extensive but nanosized interface between the pure components, which has dielectric properties different from those of either component. In the case of particulate-filled polymer nanocomposites, the specific dielectric properties have been assigned to a great extent to “interfacial regions” surrounding the nanosized particles.^[15] The interfacial regions are heterogeneous in terms of both physical and chemical structures, resulting in a particle “interaction zone” with electrical properties that are different from those of both the bulk polymer matrix and the particles.^[16] The volume (size) of the interaction zone depends not only on the intrinsic characteristics of the polymer and the nanoparticle, but also on the extrinsic mechanical, rheological, and electrical conditions.^[17–22] It is suggested that the size of the interaction zone at the polymer/nanoparticle interface is an important factor that effects the dielectric behavior of the nanocomposites.^[15,17,19]

We limit the discussion here to nanodielectric insulating nanocomposites based on polyethylene and metal-oxide nanoparticles. These show very interesting and versatile insulating and dielectric properties due to that their morphology can be tailored with specific molecular/nano-/microstructures at the interfaces.^[23] The nanoparticle size, purity and atomic termination, as well as its active surface for functionalization, which is controlled by the synthesis, are the main parameters influencing these interfaces.^[13,24–26] This molecularly controlled structure helps us to understand and reveal further the charge transport mechanisms, and other insulating materials may be approached that withstand even more severe conditions; i.e., higher electric fields, higher mechanical stresses, and higher temperatures.

Figure 1a shows an example of material development for better insulation in extruded polyethylene HVDC cables, using a bottom-up strategy: i) synthesis of high-purity metal-oxide nanoparticles, ii) surface coating of nanoparticles, and iii) nanocomposite preparation via extrusion, resulting in the formation of a “strong” interface between the polyethylene and surface-modified nanoparticles.^[9,10] Albeit this strategy seems simple and general, it relies on employing reproducible protocols with small deviations to ensure effective control of size, surface activity, and dispersion of the nanoparticles to achieve specific nanocomposite interfaces and properties. Consequently, all protocols have to be monitored by a corresponding characterization on the macro-/micro-/nanoscale (top-down strategy) to reveal the full picture of the origins of the nanodielectric behavior. So far, most reports relating to nanodielectrics deal



Amir Masoud Pourrahimi received his B.S. and M.S. degrees in polymer engineering from Amirkabir University of Technology, Tehran, Iran, in 2007 and 2009, respectively. He received his Ph.D. degree in fiber and polymer science in 2016 from KTH Royal Institute of Technology, Stockholm, Sweden. He is

currently a postdoctoral fellow at KTH. His main research interests are synthesis of inorganic nanostructures, polymer nanocomposites, nanodielectrics, and novel electrode materials for waste-water treatment.



Richard T. Olsson is an Associate Professor in the Polymeric Materials group at KTH Royal Institute of Technology, Stockholm, Sweden. He earned his Ph.D. in 2007 at KTH on microwave absorbing nanocomposites. His postdoctoral studies at CSIC (IATA), Spain, were focused on renewable nanomaterials. His main

research focus is novel nanocomposite materials and the interfaces between inorganic nanoparticles and polymers. A strong focus is in the realization of nanocomposite materials into functional prototypes.



Mikael S. Hedenqvist is currently a Professor at KTH Royal Institute of Technology, Stockholm, Sweden, where he is heading the Polymeric Materials group. He received his Ph.D. degree at the same institute 1995. He was a postdoctoral scholar in the group of Professor Richard Boyd at the University of Utah, USA. His research interests include

transport and electrical properties of polymers, polymer degradation, modeling, and biobased polymers (including protein materials).

with the material dielectric behavior on a macroscopic level and the charge mobility at interfaces has been calculated/predicted by theoretical models.^[27] Although these models are interesting as tools for extending the interpretation of measured data on a macroscopic level, the charge transport mechanisms at the finest scale (nano) of the polymer/nanoparticle interfaces is

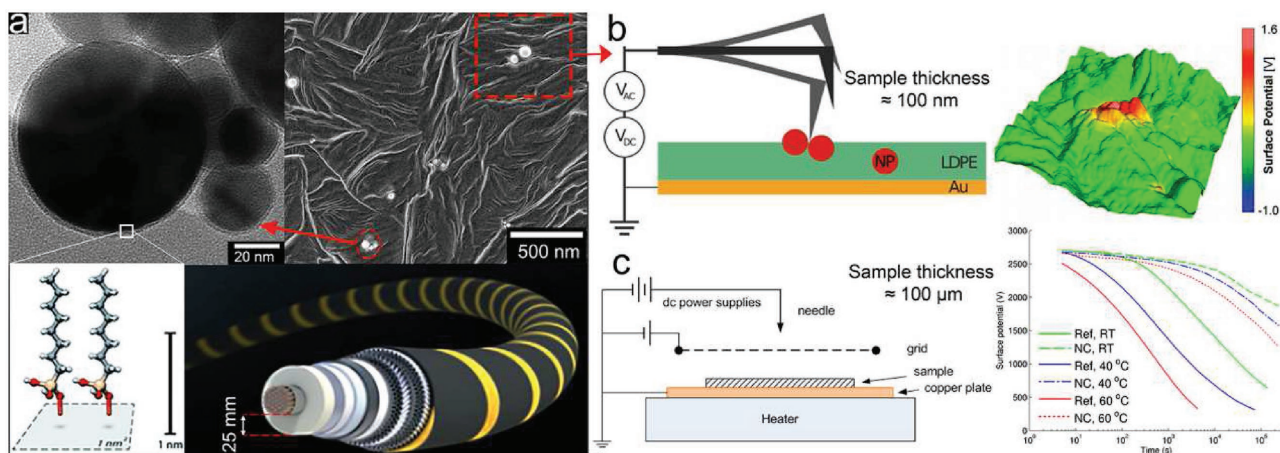


Figure 1. a) An example of insulation material development for extruded HVDC cables based on polymer nanocomposites (with TEM and SEM images); surface modification of 50 nm Al_2O_3 nanoparticles with octyltriethoxy silane (C8-) in order to be used in LDPE at 3 wt% nanoparticle loading. The TEM image is reproduced with permission.^[32] Copyright 2015, Elsevier. b) Nanoscale surface potential measurement on a film sample by ImEFM. Reproduced with permission.^[11] Copyright 2016, ACS. c) Surface potential decay at different temperatures after corona charging on LDPE (“Ref”) and LDPE/ Al_2O_3 nanocomposite (“NC”) films. Reproduced under the terms of the CC-BY 4.0 license.^[28] Copyright 2016, The Authors; published by MDPI, Basel, Switzerland.

still not fully understood, and the reduction in conductivity as a result of the addition of a low amount of insulating, semi-conductive, or even conductive nanoparticles is not completely explained. Existing models thus suffer from a limited experimental insight into the charge transport phenomenon on a nanoscale.

The surface potential measurements on a nano- and macro-scale are compared in, respectively, Figure 1b,c. The macroscopic evaluation of nanocomposites after corona charging shows a surface potential decay (SPD) with time (Figure 1c), whereas the nanoscale surface potential measurement by inter-modulation electrostatic force microscopy (ImEFM, Figure 1b) illustrates with higher resolution that a nonuniform local charge injection/extraction occurs at the polymer/nanoparticle interface, with a significant potential imbalance in the vicinity of the nanoparticles. The macroscale SPD characterization reveals that in a high electric field ($30\text{--}40\text{ kV mm}^{-1}$), the overall charge mobility of a low-density polyethylene (LDPE) sample is reduced from 4×10^{-14} to $5 \times 10^{-15}\text{ m}^2\text{ V}^{-1}\text{ s}^{-1}$ at 60°C by the addition of 3 wt% Al_2O_3 nanoparticles.^[28] It has been suggested that reduction in charge mobility in LDPE/ Al_2O_3 nanocomposites, referred to as “charge suppression,” is due to an increased barrier against tunneling/hopping of charges,^[23] which also leads to a reduction in DC conductivity of the polyethylene matrix (from 2×10^{-14} to 10^{-15} S m^{-1}).^[28] Mapping of the charge mobility on a nanometer scale can here provide an important tool for understanding of how local charge injection/extraction influences the macroscopic electrical measurements of nanodielectrics (revealing charge-transport mechanisms).^[11] More experimental data on charge mobility at polymer/nanoparticle interfaces (nanoscale) may clarify better the interpretation of electrical data on a macroscale.

A number of observations have been made in the last few years in the field of nanodielectrics, and these have been assessed by multiscale characterization. These observations are as follows:

- (1) In order to reduce the charge mobility or DC conductivity for composites based on noninsulating particles, the particle volume fraction in the composite system should typically be lower than the percolation threshold, where conduction may occur by tunneling between particles due to a short interparticle distance.^[29,30] At this critical interparticle distance the trapping of charge carriers by noninsulating particles is not anymore beneficial for DC conductivity reduction. The percolation threshold and the corresponding interparticle distance are not fixed values, and could be altered with different intrinsic nanocomposite components properties and external conditions. Hong et al.^[29] showed that the conductivity of LDPE/ ZnO nano- and microcomposites increased sharply at a critical ZnO interparticle distance of 40 nm, whereas Tjong and Liang^[31] determined 400 nm as a critical interparticle distance for similar composite systems. This inconsistency indicates that the determination of this value needs more definitive experimental methods, nanocomposite components characteristics, and extensive micrograph analysis for particle dispersion determination.
- (2) The larger surface area of smaller particles/agglomerates provides a large specific interface/interaction zone, and this may improve the insulating properties.^[9,15,29] It should be noted that while the high surface area of the particles may provide more trapping sites for charge carriers, it is important that new charge carriers are not introduced with the addition of particles (refer to Equation (1)).
- (3) A surface modification of hydrophilic metal-oxide nanoparticles is required in order to improve their dispersion in the hydrophobic polyethylene matrix. The surface modification may also enhance the interfacial strength, and can lead to superior electrical, mechanical and thermal properties of the nanocomposites.^[9,10,12,32,33] Although most of the literature reports have stated that the surface modification of the nanoparticles leads to better insulating properties of the nanocomposites, but there are few experiments to assess

the exact surface chemistry, the size and the porosity of the coatings. The discussion about the charge transport behavior at the interface without determination of these parameters could be speculative and questionable.

- (4) The polyethylene morphology is occasionally altered by the addition of particles, and polymer chains may be constrained at the particle interface.^[34] The nanoparticles may also act as nucleating agents, which change the polyethylene microstructure and crystallization behavior.^[9] This feature needs to be addressed further.

To explain the above observations, a number of theories have been put forward in order to explain the underlying charge transport mechanisms at interfaces in nanocomposite-based insulation materials, including the bipolar model (discussed in more detail in a later section):

- (1) “Lewis’s model”: While the polymer and nanoparticle phases have dielectric properties with a specific and constant intensity; the interface/interphase has intermediate dielectric properties, gradually changing from one phase to the other.^[17]
- (2) “Tanaka’s model” (multicore model): An electric double-layer or multilayer is formed around the nanoparticle having different charge characteristics.^[17,35] The layer(s) adjacent to the nanoparticle surface have a high charge density and contain trapped charge carriers. The next layers contain mobile charge carriers, which can migrate toward the nanoparticle surface.
- (3) “Multiregion structure model”: This model is a more recent version of the multicore model, and considers that the interfacial region around the spherical particle in the polymer nanocomposites contains three different subregions.^[36] A bonded region, rich in deep traps, is situated next to the particle surface and connected to this by both primary or secondary bonds. A transitional region is situated next to the bonded layer with properties intermediate between the bonded and normal (the third, polymer bulk) region, the latter containing the “unaffected” polymer matrix. Deep traps are considered to decrease outward from the particle, being replaced by shallow traps.
- (4) The “induced dipole moment model”. Whereas “chemical defects,” such as carbonyl groups, are permanent dipoles, the nanoparticles become induced dipoles under a high electric field.^[37] Both types of polarization/dipoles lead to more traps for electric charge carriers. The size of the dipole moment, trapping density and depth, depend on the dielectric constant, size, and shape of the nanoparticles.

In addition, the detailed chemistry at the particle surface deserves special attention. The hydroxyl group density at the surface of metal-oxide nanoparticles which are able to adsorb charge carriers^[33,38] may possibly affect the overall charge carrier concentration and mobility in polymer matrix, regardless of the nanoparticle’s bulk dielectric properties. The hydrophilic surface-active hydroxylated nanoparticles can adsorb water when the nanocomposite is exposed to moisture. It has been shown that a prolonged immersion in water has little effect on the unfilled polyethylene, whereas an accumulation of water was detected at nanoparticle/matrix interfaces in

nanocomposite systems.^[39,40] It was suggested that the water present around the nanoparticle/aggregate forms a diffuse structure similar to Tanaka’s model, i.e., a water shell model.^[38] The shell was proposed to consist of water in clusters of various sizes, increasing with moisture conditioning time. The water shell is formed by former water layer through associating of water molecules to hydroxyl groups at the nanoparticles surface, and then this water layer can continue attracting more water molecules.^[40] Most studies on the effects of water on the electrical performance of nanocomposites considers that the water shells can percolate at a larger particle content and finally lead to a poor insulating material with a large amount of heterocharge formation and decreased breakdown strength.^[39–42] On the other hand, Lei and Sheng^[43] indicated that water can act as a plasticizer in polyethylene, thus increasing the free volume and the amount of trapping centers, which possibly reduces the charge carrier injection. It must be stated here that in order for water to plasticize polyethylene, or any other polymer for that matter, the water content must be quite high (on the order of several percent). To conclude, the effect of water on the electrical breakdown behavior of nanocomposites is complex and needs to be explored further.^[44]

The information available is not able to completely explain the mechanisms of charge transport at nanometric interfaces in insulating nanocomposites. The theoretical models can today interpret parts of the charge transport mechanisms based on pure components properties and macroscale characterization of the nanodielectrics/nanocomposites. The theories for predicting the behavior of nanodielectric materials have been improved since the Lewis model in 1994, but they are still limited in revealing the charge transport mechanisms in specific nanocomposite insulation materials under different external conditions, i.e., variations in the applied electric field. Some of these limitations are due to the fact that some of the above models were first derived for noninsulating materials; i.e., semiconductive materials, colloids, and varistor ceramics in electronic applications.^[17] In addition, the analytical models primarily refer to specific case studies or restrictive conditions, which are far from the real conditions of high-voltage insulation materials.^[45] An understanding of the impact of the detailed surface chemistry of the nanoparticles used (hydroxyl groups and other functionalities, which are able to trap charges via hydrogen bonds) has to some extent improved the models.^[33,38] Nevertheless, the application and development of these models under real conditions for nanocomposite insulation materials, are still difficult, due to a lack of experimental data on the nature of charge carriers and traps at the interfaces. Hence, it is important to improve the “theoretical models” for nanodielectric performance with further experimental data.

Here, we summarize the status of material development in the field of HVDC (mainly polyethylene) insulation materials, and present an overall picture of the characteristics of the nanocomposite phases (polyethylene, metal-oxide nanoparticles, and interfaces). We also discuss some cases which are not consistent with accepted ideas and theoretical models, which raise questions that cannot be completely answered by existing models. We briefly review the existing models and attempt to evaluate and discuss the performance of polyethylene nanocomposite insulation materials with both theoretical models and recent

nanoscopic experiments to reveal the charge transport mechanisms at the interfaces. This approach, i.e., investigating the charge transport mechanisms in highly insulating nanocomposites from the molecular to the macro level, will hopefully contribute to achieve the goal of having insulation materials with properties that can withstand 1 MV levels by not later than 2030.

2. Polyethylene Nanocomposites for High Voltage Insulation

2.1. Low-Density Polyethylene

The polyethylene matrix itself can be considered as a nanodielectric.^[23] Polyethylene is the simplest hydrocarbon polymer and was first produced by the polymerization of ethylene under high pressure (1000 atm) in 1933 by Eric William Fawcett and Reginald Oswald Gibson.^[46] Since then, it has undergone continuous development to meet different needs. Despite its simple repeating unit, the physical structure of polyethylene (chain conformation, crystal–amorphous interface, crystal size and shape, and supermolecular structure) is complex,^[47] but polyethylene has nevertheless been regarded as a model polymer for semicrystalline polymers in general.^[48] The most common grades of polyethylene are low-density polyethylene (LDPE), high-density polyethylene (HDPE), and linear low-density polyethylene (LLDPE). These vary in molecular weight, molecular weight distribution and degree of branching (short chain branching (SCB), and long chain branching (LCB)), which control the final properties, including crystallinity/density and mechanical properties.^[49] The SCB prevents polyethylene chains from forming thick crystals since the branching points cannot fit in the crystals. The LCBs increase the free volume and decrease the critical shear rate for the onset of shear thinning.^[50] The free volume is the remaining volume when the incompressible or hardcore volume is subtracted from the specific volume; which can be “squeezed out” by going down to 0 K and or applying a very high pressure. The polymer chain mobility increases with increasing free volume.^[51] LDPE contains both LCBs (at least 1–2 per 1000 carbons) and SCBs (10–50 per 1000 carbons).^[52] This level of branching in LDPE leads to remarkably strong shear thinning and a low melt viscosity during extrusion, which results in good processability, e.g., in the case of extruded HVDC-cable insulation materials. LDPE also meets other requirements such as long-term mechanical and chemical stability, high electrical breakdown strength, and high DC resistivity.^[8] The latter, which currently gives the lowest possible transmission loss for long distances, makes LDPE the best candidate for the insulating part of extruded HVDC cables.

In order to understand the beneficial electrical properties of polyethylene, its semicrystalline and electronic structures on different scales (from micro to molecular level) are illustrated in **Figure 2**. Melt-crystallized polyethylene forms micrometer-sized polycrystalline spherulites on cooling, and these grow until they meet neighboring spherulites, yielding interspherulitic interphases/boundaries.^[49,53] The amount of spherulites and their interphases per volume increases with decreasing spherulite size, which can be controlled by the use of nucleating agents.

The morphology on a spherulitic level has been reported to have an effect on the electrical properties of polyethylene; the addition of sorbitol (as a nucleating agent, yielding smaller spherulites) to LDPE suppressed the charge accumulation.^[54] Low molecular weight polyethylene chains possibly accumulate at the spherulitic boundaries and form mainly low-crystalline interphases.^[47] A hypothesis is that these interphases provide new charge carrier traps, in addition to those present within the semicrystalline spherulites, and hence the charge mobility may decrease even more.^[55] Determining purely the effects of spherulite sizes and boundaries is not however straightforward, since other parameters are also changed when the spherulite size is changed, including crystal size and size distributions. Hence, this hypothesis needs to be proved using systems where the spherulite effect is isolated from all other parameters. One way of approaching this condition is by using polyethylene chains with regular intervals of chain branches, minimizing variations in crystal thickness when external parameters are varied.^[56]

On a sub-micrometer scale, each spherulite contains stacks of crystalline lamellae sandwiching amorphous layers. The radial (growth) direction of the spherulites is close to the lateral direction of the crystal lamellae, i.e., along the crystallographic *b* axis.^[47] Polyethylene chain segments appear as chain ends, tie chains, and trapped entanglements/loops in the amorphous regions,^[57] whereas chain segments in the crystalline/amorphous interface mainly occur as “chain folds” parallel to the (110) crystallographic plane.^[47] The chains in the crystalline phase are oriented along the *c*-axis (chain axis) and are usually packed in 10–20 nm thick crystalline lamellae. The polyethylene chains with a higher molecular weight crystallize at higher temperature or at shorter times in thicker “dominant” crystals, while low molecular weight species lead to thinner “subsidiary” crystals.^[47] The extent of this type of segregated structure depends on the molecular architecture and crystallization kinetics, and it seems to affect the electrical properties. The crystallization of polyethylene can be recorded with differential scanning calorimetry (DSC) in order to find the crystallinity and crystal thickness. The mass crystallinity (w_c) of the polymer is determined according to the total enthalpy method:^[58]

$$w_c = \frac{\Delta H_f}{w_p \left(\Delta H_f^0 - \int_{T_1}^{T_m^0} (c_{p,a} - c_{p,c}) dT \right)} \quad (2)$$

where ΔH_f is the measured melting enthalpy, ΔH_f^0 is the melting enthalpy for 100% crystalline polyethylene (293 J g^{−1})^[58] at the equilibrium melting point, T_1 is an arbitrary temperature below the melting range, T_m^0 is the equilibrium melting temperature (141.5 °C), w_p is the mass fraction of polymer in the polymer/particle composite system,^[32] and $c_{p,a}$ and $c_{p,c}$ are, respectively, the specific heat capacities of the amorphous and crystalline components, which are obtained from Wunderlich and Baur.^[59] The crystal thickness (L_c) associated with the melting peak temperature is calculated according to the Thomson–Gibbs equation:^[47]

$$L_c = \frac{2\sigma_e}{\Delta H_f^0 \rho_c \left(1 - \frac{T_m}{T_m^0} \right)} \quad (3)$$

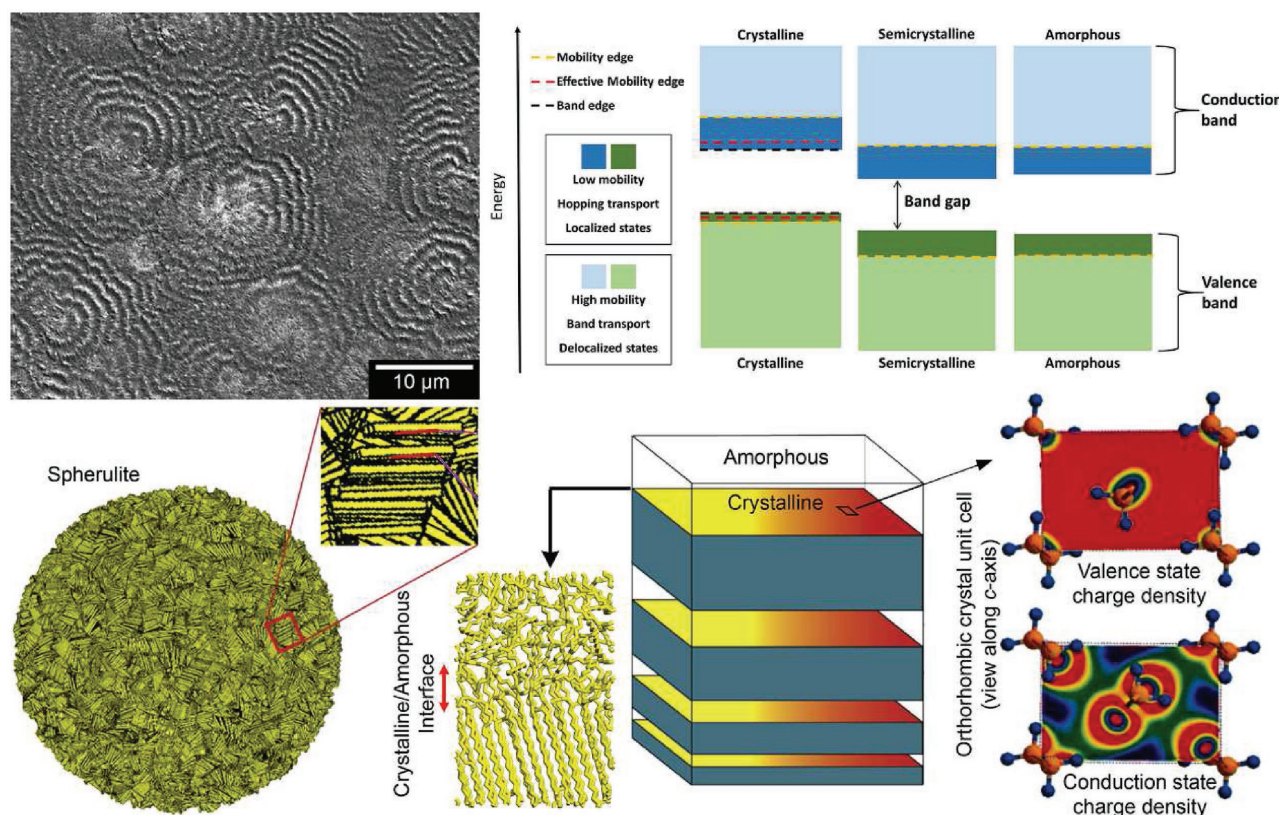


Figure 2. The semicrystalline structure of polyethylene together with its electronic structure. An SEM image of micrometer-sized spherulites is shown in the upper left. Reproduced with permission.^[53] Copyright 2011, The Society of Plastic Engineers. The lower left and middle figures show the crystal lamellae and amorphous layer in the spherulite and the molecular features of the crystal/interface/amorphous layers. Reproduced with permission.^[57] Copyright 2015, Elsevier. The lower right image illustrates the charge density in the crystal unit cell at different conduction and valence states, increasing from red to indigo, similar to the order of colors in a rainbow. Reproduced with permission.^[67] Copyright 2000, American Physical Society. The electronic band structure of semicrystalline polyethylene is shown at the upper right. Reproduced under the terms of the CC-BY 4.0 license.^[68] Copyright 2017 The Authors; published by American Institute of Physics.

where T_m is the melting peak temperature, $\rho_c = 1003.0 \text{ kg m}^{-3}$ and $\rho_a = 851.9 \text{ kg m}^{-3}$ are the densities of, respectively, the crystalline and amorphous components, and $\sigma_e = 93 \text{ mJ m}^{-2}$ is the fold surface free energy for linear polyethylene.^[60,61] **Figure 3a** shows the melting thermograms of pristine LDPE and HDPE together with those of their blends. The melting points of pristine LDPE and HDPE are 110 and 133 °C, respectively, which corresponded to average crystal thicknesses of 8.5 and 30 nm based on Equation (3). By adding HDPE into LDPE, there is no noticeable change in the melting point or crystal thickness of the LDPE component, while a lower melting point and corresponding smaller crystal thickness is observed for the HDPE component (Figures 3a,b). The most important feature in the LDPE/HDPE blends is that the addition of 1–20 wt% HDPE to LDPE introduces a small population of homogeneously distributed new cocrystals with intermediate thicknesses, while the total crystallinity is not changed (Figures 3a–c.^[62–64]). The distribution of these thicker cocrystals within the semicrystalline LDPE phase was also detected by SEM (Figures 3f–i). The presence of thicker crystals in the blends with 5 wt% HDPE reduced the DC conductivity of LDPE by approximately one order of magnitude to about $10^{-15} \text{ S m}^{-1}$ in high electric fields (30 kV mm^{-1}) and at a high temperature (70 °C), Figure 3d.

The DC conductivity measurements are usually performed following a standard procedure^[65] by applying direct current (DC) voltage over the film sample and measuring the charging current via an electrometer.^[13] Although an over-voltage protection secures the electrometer from damaging due to possible overshoots and a low-pass filter removes high frequency disturbance, there are still some noises observed in the charging current curve indicating an uncertainty of this macroscale characterization. Therefore, it is suggested that this method should be combined with other dielectric macroscale characterizations. The authors, although providing a clear full-picture of the LDPE/HDPE microstructure,^[64] the mechanism(s) behind the reduction in DC conductivity in the presence of HDPE was not fully revealed.^[62] HDPE has, as compared to LDPE, a lower permeation rate of charge carriers (ionic and polar molecules) due to a higher crystallinity, which leads to lower charge mobility and DC conductivity (Equation (1)).^[8,66] In the blends presented with a low HDPE content, the crystallinity was, however, not changed significantly. It is questionable to draw general conclusions regarding the charge mobility behavior in polyethylene blends without considering detailed blend structure (at nano and molecular scales).

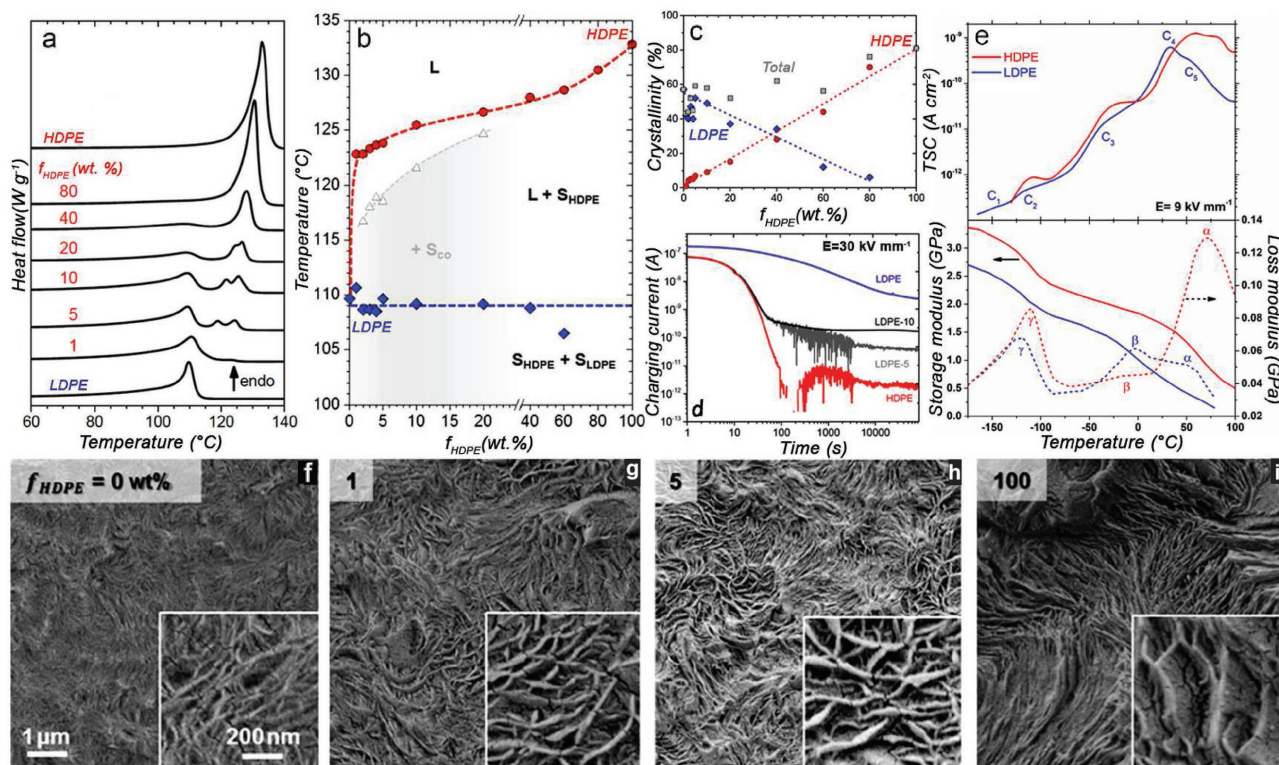


Figure 3. The crystallization, rheological and electrical properties of LDPE, HDPE, and their blends: a) DSC melting thermogram, b) corresponding temperature–composition diagram (L, S, and “co” stand for, respectively, the molten state, solid state, and cocrystals), c) crystallinity, and d) charging current of LDPE/HDPE blends (the number refers to the mass content of HDPE). e) X-ray induced TSC spectra of LDPE and HDPE together with DMA results of LDPE and HDPE (temperature scan at a bending frequency of 1 Hz), a–c, f–i) Reproduced with permission.^[64] d) Reproduced with permission.^[63] Copyright 2017, M. Andersson. e) Data replotted from ref. [72] (top) and from ref. [73] (bottom).

On a molecular level, the most commonly observed crystal unit cell for polyethylene is the base-centered orthorhombic unit cell, with lattice constants: $a = 0.74$ nm, $b = 0.495$ nm, and $c = 0.253$ nm.^[47] The electronic charge associated with the valence band is mainly around the C–C and C–H bonds (intrachain), while that associated with the conduction band is located between neighboring chains (interchain) (Figure 2, lower right).^[67] The energy gap between the valence and conduction bands for semicrystalline polyethylene is 8.8 eV, which is ≈ 0.65 eV above the vacuum level, showing that polyethylene is an electrically insulating material.^[23]

It is commonly believed that polyethylene, with its semicrystalline structure, has relatively conductive crystalline lamellae, separated by resistive amorphous regions,^[23] where the amorphous phase produces a tunneling potential barrier between the donors and acceptors of crystal lamellae, suppressing charge accumulation and reducing the conductivity with reference to the crystalline phase.^[23] Density-functional-theory (DFT) calculations, combined with molecular dynamic (MD) simulations, can yield realistic electronic structures and electrical properties using the detailed molecular structure of polyethylene. DFT/MD reveals that the pure crystalline phase has a slightly larger bandgap than the pure amorphous phase (Figure 2, upper right). The calculated bandgaps for amorphous, crystalline and semicrystalline structures are, respectively, 5.89, 6.00, and 5.82 eV,^[68] and their calculated charge mobilities at room temperature are, respectively, 0.007, 0.008,

and $0.0002 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.^[69] The valence and conduction band edges move closer together when the crystalline and amorphous phases are considered as parts of an overall semicrystalline phase. Also Chen et al.^[70] calculated a bandgap of the semicrystalline phase that was lower than the pure crystal and amorphous bandgaps. Despite its smaller bandgap, the charge mobility of the semicrystalline phase is at least one order of magnitude lower than that of the pure crystalline and amorphous phases, which means that the semicrystalline phase has the lowest apparent conductivity of these three structures. These calculations were however made on a periodic boundary conditions (simulation box was set as an orthorhombic ($4a$, $6b$, $8c$) polyethylene crystal vertically aligned with a 2.462 nm thick amorphous layer) and a physically relaxed/equilibrated semicrystalline structure without any applied electric field.^[68] The local conformational disorder of polyethylene chain segments in the amorphous phase may also produce localized states or shallow traps at the edge of both conduction and valence bands, with a depth in the range 0.15–0.3 eV and a residence time of $\approx 10^{-12}$ s.^[71] In addition to the shallow traps, chemical defects (disorders) like hydroxyl or ketone functions or double bonds may create deeper traps for electron and holes (0.04–1.5 eV) and virtually infinite residence times.^[71] The charge transport mechanisms are discussed in details furthermore in the last section. Recently Chen et al.^[70] calculated trap depths for electrons and holes in polyethylene having different chemical (e.g., double bonds and polar groups) and physical (e.g., kinks,

chain branches and density variations) defects/distortions. The computed trap depths were 0–2.8 eV (electrons) and 0.4–3.4 eV (holes) for chemical defects, and 0.07–1.4 eV (electrons) and 0–1.09 eV (holes) for physical defects. The computed electron trap depth increased with increasing material density, which was interpreted as that the electrons prefer to reside in lower density regions.

The charge trap depth, which controls the charge mobility, can be experimentally estimated by X-ray-induced thermally simulated currents (TSC) and thermoluminescence (TL).^[72] The X-ray irradiation induces free electrons (or holes) in the sample, which are captured by traps at a low temperature. Subsequently, the thermal release of electrons (or holes) from traps results in TSC under a biased field and increasing temperature. The TL spectra are obtained by the recombination of holes and released electrons. The presence of deep traps due to chemical defects can also be revealed with X-ray-induced TSC.^[72] The analyses of X-ray-induced TSC and TL can give information about the nature of the traps, including their depth and origin. Figure 3e shows that the typical TSC spectrum of LDPE has five peaks (C_1 – C_5) in the temperature range from –160 to 80 °C.^[72] The C_2 , C_3 , and C_5 peaks (at $\approx$ –90, –20, and 60 °C) correspond, respectively, to the γ , β , and α relaxations (as revealed by dynamic mechanical analysis (DMA));^[73] showing that detrapping is favored by increasing molecular mobility. The C_4 peak, appearing between –40 and 40 °C, originates from traps with a depth of 1.0–1.4 eV, which are believed to be located at the amorphous/crystalline interface,^[72] and may be due to the local chain fold pattern. However, the latter hypothesis cannot be experimentally observed by this method, since the chemical defects also produce similar trap depths, and it is difficult to distinguish between the traps due to the amorphous/crystalline interface and chemical defects. X-ray-induced TSC and TL spectra for HDPE shows that the C_5 TSC peak originates from traps with a depth of 1.7 eV, which is also observed at a higher temperature for HDPE than for LDPE (Figure 3e).

The picture of the origin of the electric properties of polyethylene under real HVDC insulation conditions is still incomplete and the experimentally observed bandgap of polyethylene has not been fully predicted by modeling/simulations. In addition, the calculated bandgap structures of the crystalline, amorphous and interfacial components of polyethylene cannot be verified directly by any experiment. There is no experimental technique that can detect the electric properties of these individual components under an applied high electric field. Bearing in mind the complexity of the electrical properties of polyethylene itself, it is a challenge to be able to interpret fully the electrical effects of an additional component (nanoparticle) on the electric properties of a polyethylene nanocomposite. Nevertheless, we attempt to interpret the measured electrical properties of polyethylene nanocomposites in the following chapters by considering the presence of a polyethylene/nanoparticle interface.

2.2. Functionalized Metal-Oxide Nanoparticles

A range of different nanoparticles have been studied to improve the electrical insulating properties of polyethylene, including both insulating^[10–13,28] and semiconductive^[9,29,74,75] metal-oxide

nanoparticles, as well as conductive metal and carbon-based nanoparticles.^[76,77] Metal-oxide particles prepared by aqueous condensation are interesting alternatives as functional fillers in the next generation of nanodielectrics. The classical “chimie-douce” enforced precipitation reactions meet the fundamental requirement of having particles with a known preparation history, that to the greatest possible extent can exclude contaminations from the electrical analysis. These precipitation reactions also provide versatility, and make possible a rational and systematic study of the effect of reaction parameters on the particle morphology. Post-synthesis protocols are required for the removal of counter-ion residues (synthesis by-products) from the surface of the precipitated nano-/micro-sized metal-oxide particles,^[24–26] in order to be able to relate the nanocomposite electrical performance to properties of the particles, e.g., size, size distributions, specific crystal habit, atomic terminations of surfaces, and inherent electrical characteristics. The presence of counter-ion residues on the surface of the particles leads not only to particle sedimentation and low colloidal stability in an aqueous medium, but also to further deterioration of surface functionalization in order to be compatible with the hydrophobic LDPE.^[9,12,78] Besides the surface passivation effect, the counter-ion residues lead to formation of heterocharge carriers in the LDPE matrix, and this results in earlier electrical breakdown of the nanocomposites. Novel postsynthesis cleaning methods based on ultrasonication (removal of intra-aggregate counter-ions and increasing the colloidal stability) and heat-treatment (ultimate removal of counter-ions from surface charged particles) are suggested in order to achieve ultrapure metal-oxide nanoparticles for high-voltage insulation materials.^[9,10,24–26] Dynamic light scattering may provide an efficient tool for monitoring the removal of “counter-ion impurities” and colloidal stability after the aqueous synthesis reactions, in combination with particle characterization by X-ray photoelectron spectroscopy for high precision detection of surface counter-ions, X-ray diffraction (phase and crystal habit), microscopy (size distributions and morphology) and Brunauer–Emmett–Teller method (BET) for surface area and porosity determinations.^[13,24–26,78]

Figure 4a shows ZnO nano-/micro-sized particle surfaces dominantly terminated by either zinc or oxygen, as a result of reaction conditions favoring the exposure of specific crystal planes. Different “atomic terminations” of crystal surfaces, illustrated in Figure 4b, have previously been shown to result in marked differences in properties such as photocatalytic and antimicrobial activity.^[24,79] Refined particle surfaces may affect the ability to grow certain coatings on the surfaces of the particles and to integrate functionalities that present electronic substructures that not only affect electron distribution but can also adsorb migrating species (i.e., that remain in the polymer matrix as a result of the polymerization/crosslinking reactions),^[33,80] in addition to their compatibility with and dispersion in the polyethylene. Figure 4c shows an example of a covalently bonded silane coating (silsesquioxane) which have been extensively explored for their functionalities in the search for compatibility between nanoparticles and polymer matrices.^[9,12,78,81,82] However, although silsesquioxane coatings are known to successfully provide good particle dispersion in many systems, the manner in which the porosity of the coatings affects the “charge

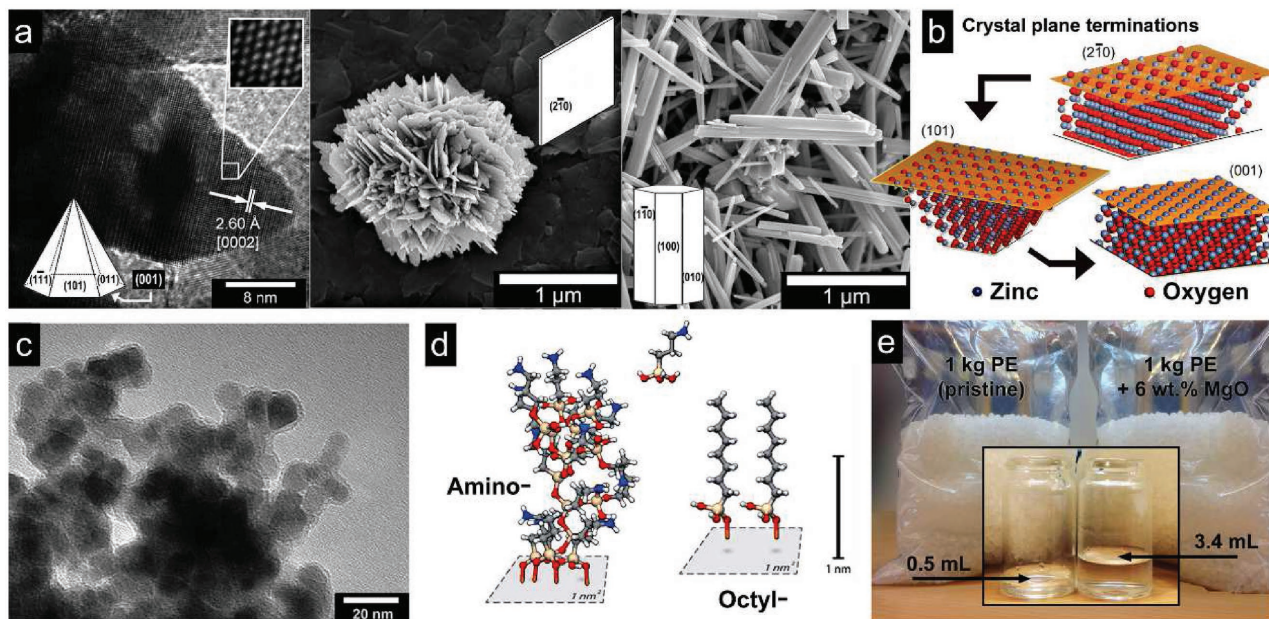


Figure 4. The structure of functionalized metal-oxide nano-/microparticles from molecular to microstructure; a) TEM/SEM images of different ZnO nanostructures with different crystal plane terminations. Reproduced with permission.^[79] Copyright 2017, Elsevier. b) Crystal model of different planar terminations. c) TEM image of octyltriethoxy silsesquioxane (C8-) coated Fe₃O₄ nanoparticles. Reproduced with permission.^[78] Copyright 2015, The Royal Society of Chemistry. d) Schematic drawings of the amino- and octyl-based silsesquioxane coatings, where gold, gray, red, blue, and white spheres correspond, respectively, to silicon, carbon, oxygen, nitrogen, and hydrogen atoms. Reproduced with permission.^[78] Copyright 2015, The Royal Society of Chemistry. e) Illustration of water present in pristine polyethylene and in a polyethylene nanocomposite at RH 50%.

trapping" phenomenon has only recently been investigated.^[9,12] The porosity of the coating stems from the existence of competing condensation reactions that occur between the silanol groups of molecules to be deposited (before reaching the surface of the particles) and the directly deposited individual molecules of the surface modification medium (Figure 4d). These reactions result in silsesquioxane coatings deposited as clusters with embedded functionalities which prevent the formation of completely condensed silsesquioxane layers in the vicinity of the particle surface.^[78] Selected particle fillers may serve different purposes in a dielectric nanocomposite, which is discussed in more detail in Section 4.1. It is difficult to assess the effect of the porosity of covalently bonded silane coatings since even a hydrophobic coating inevitably introduces water into the nanocomposite system. Figure 4e shows the quantity of water in polyethylene with and without nanoparticles. A nanocomposite with 6 wt% octyl-functional silsesquioxane-coated magnesium oxide nanoparticles (C8-MgO) contains almost 7 times more water than the pristine polyethylene. The DC conductivity for this system is approximately an order of magnitude lower when examining short term data (10 s) as compared to the particle-free material despite the high water content.^[44] However, at a longer period of time (>2 h) the nanocomposites show no improvement in the insulating properties as compared to unfilled LDPE. Hence, the effect of water seems complex and needs further investigations to reveal fully the behavior of the insulation in the presence of water. In practice, HVDC cables contain a metal barrier layer to prevent the uptake of water, but it is not unlikely that water is present in the material before the "encapsulation." It is also well documented that the nature of the particle surface has a direct impact on the crystallization of

polyethylene (Figure 5g),^[9,10] emphasizing the need to develop refined and predictable protocols that not only provide a long-term particle-core protection in cable insulation,^[10] but also meet potential future challenges from a surface chemical perspective in different applications.^[83–85]

2.3. Interfaces in LDPE/Metal-Oxide Nanocomposites

The pristine semicrystalline LDPE contains microscopic interfaces between spherulites and nanoscopic interfaces between the crystalline and amorphous phases; but, new types of interface develop between the LDPE matrix and the nanoparticles in the nanocomposite systems. The latter interfaces are considered to play a dominant role for the nanocomposite properties. However, some composite systems have properties that cannot be explained only by the presence of polymer/nanoparticle interfaces. Various metal-oxide nanoparticles (both semiconductive: ZnO and insulating: MgO and Al₂O₃) leads to a dramatic reduction in DC conductivity compared to the reference LDPE.

Figure 5a shows the calculated normalized interface-to-polymer-matrix volume in composite systems as a function of particle volume fraction for ZnO particles of different size (the ratio is 1 when the interface occupies the whole polymer matrix volume).^[15] Figure 5a (inset) illustrates the hypothetical interaction zone formed around micro- and nanosized particles regardless of the exact nature of the interaction zone (the particles are not shown to scale). It is assumed that the thickness (not volume) of the interaction zone is independent of particle size. The interaction-zone volume formed by nanoparticles dominates the composite volume at a much lower particle loading than that

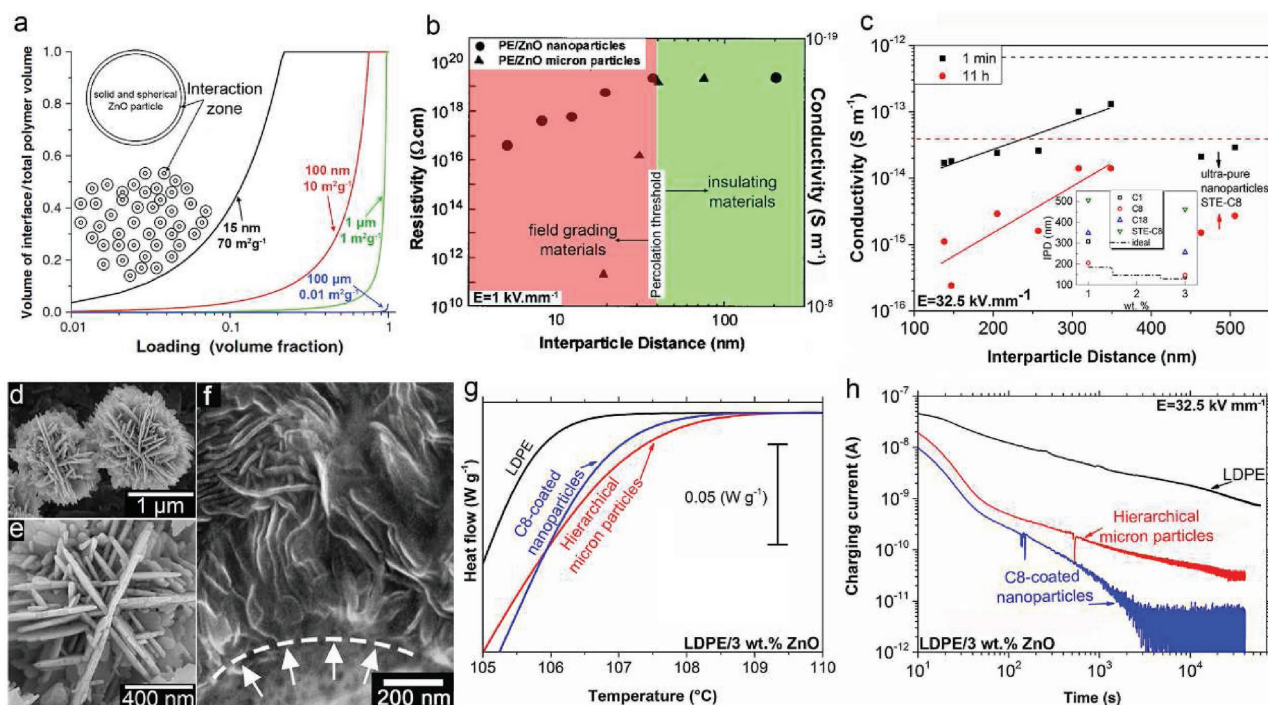


Figure 5. a) Interface-to-matrix volume ratios of composites based on ZnO particles with different sizes and specific-surface-areas as a function of particle loading. b) The DC conductivity/resistivity plotted against interparticle distance for LDPE composites based on nano- and micrometer-sized ZnO particles. Reproduced with permission.^[29] Copyright 2003, AIP. c) DC conductivity of LDPE containing surface modified ZnO particles as a function of interparticle distance. The black and red dashed lines correspond to the conductivity values of unfilled LDPE after 1 min and 11 h, respectively. The inset shows the interparticle distance versus the weight fraction of 25 nm ZnO nanoparticles with different surface functionalities used in the nanocomposites (the black dash line shows the theoretical interparticle distance). Reproduced with permission.^[9] d–f) SEM images of hierarchical micron-sized ZnO and the corresponding LDPE composite after permanganic acid etching (the arrows show the radial direction of a ZnO particle at its outer surface). d,e) Reproduced with permission.^[24] Copyright 2016, American Chemical Society. f) Reproduced with permission.^[9] g,h) DSC crystallization thermograms (g) and charging current (h) of composites based on ZnO nano- and hierarchical particles. Reproduced with permission.^[9]

of the microparticles. The amount of interface is related to the specific surface area of the particles. Assuming solid spherical particles, the specific surface area (SSA; $\text{m}^2 \text{g}^{-1}$) is obtained as:

$$\text{SSA} = \frac{6000}{\rho d} \quad (4)$$

where ρ is the particle bulk density (g cm^{-3}) and d is the particle diameter (μm) (the calculation of the SSA in Figure 5a was based on ZnO). It is evident that by breaking micro-sized particles (aggregates) into individual nanoparticles in the nanocomposite system leads to a huge interfacial surface area that can interact directly with the polymer matrix. As can be seen in Figure 5a, the interface volume starts to occupy the whole polymer matrix volume at a specific loading, which depends on the particle size (or its SSA). The observed differences in this level of loading (referred to as the “percolation threshold”) can be explained by considering the interparticle distance (IPD). For a face-centered cubic lattice arrangement of spherical nanoparticles in the polymer matrix, the theoretical surface-to-surface IPD, for a given particle size (d) and volume fraction of particles (V_f), in the composite system is calculated by:^[20]

$$\text{IPD} = d \left(\frac{\pi^{1/3}}{6V_f} - 1 \right) \quad (5)$$

Figure 5b shows the DC resistivity/conductivity plotted against the calculated interparticle distance with an applied electric field of $E = 1 \text{ kV mm}^{-1}$.^[29] A critical interparticle distance of $\approx 40 \text{ nm}$ is estimated to be the interparticle percolation distance for LDPE/ZnO nano- and microcomposites, although conduction between particles may commence via tunneling leading to a dramatic increase in DC conductivity for shorter interparticle distances. In another study, the conductivity of LDPE/ZnO nano and microcomposites was measured by an impedance analyzer at a frequency of 50 Hz, which yielded a percolation interparticle distance of 400 nm.^[31] Although the results from different measurements (employing DC and AC) cannot be compared, this large difference for critical IPD is unexpected. In both papers,^[29,31] the IPD was theoretically obtained from Equation (5), however the IPD calculated from examination of several scanning electron microscopy images (image analysis) was probably more reliable. The critical interparticle distance in both papers was similar for both the nanocomposite and microcomposite systems, suggesting that this may be an intrinsic polymer matrix property; independent of particle size and external conditions. However, the latter independence is still controversial in the case of conductive and semiconductive filler particles. This matter can be due to nonlinearity, defined as the slope of the current–voltage curve on a log scale (conductivity–electric field), which is observed

in field-grading materials. The LDPE shows no noticeable nonlinearity in the 0.1–2 kV mm⁻¹ electric field range, but the LDPE nanocomposite shows increasing nonlinearity with increasing ZnO nanoparticle loading. At a volume fraction of about 0.04 the nonlinearity is even higher than for pure ZnO nanoparticles.^[30] This nonlinearity may increase at even higher electric fields, resulting in relatively conductive rather than insulating materials. This suggests that if conductive and semiconductive nanoparticles are to be used in HVDC insulating nanocomposites, they have to be added in very low concentrations.

Figure 5c shows the electrical insulation capacity of LDPE nanocomposites, based on surface-modified 25 nm ZnO nanoparticles, at ultralow loadings (<3 wt% or 0.5 vol%) under a DC electric field of 32.5 kV mm⁻¹.^[9] This particular electric field corresponds to the real condition of HVDC cables with a transmission voltage of 800 kV for an insulation thickness of 25 mm (Figure 1a). The surfaces of the ZnO nanoparticles were modified using a series of silanes with terminal alkyl groups of different lengths (methyl (C1), octyl (C8), and octadecyl (C18) groups). The necessity of surface coating hydrophilic metal-oxide nanoparticles for use in a hydrophobic polyethylene matrix is discussed in Section 2.2. The composites based on uncoated ZnO nanoparticles show an early electrical breakdown under a high electric field (32.5 kV mm⁻¹), due likely to the presence of large agglomerates (1–50 µm). No large agglomerates were detected in the surface-modified nanocomposites, and there was a dramatic reduction in conductivity for the polyethylene (Figure 5c). The dispersion of the modified ZnO nanoparticles in the LDPE matrix was assessed by comparing the IPD obtained from an examination of scanning electron microscopy images with its theoretical (ideal) value (Equation (5)). The theoretical value indicates a perfect dispersion of the nanoparticles in the LDPE matrix. Figure 5c (inset) shows that the smallest IPD was observed for the LDPE nanocomposite based on 3 wt% C8-coated ZnO nanoparticles (147 nm), which is close to the theoretical value (129 nm), indicating perfect dispersion.^[9] These nanocomposites also showed a conductivity (2.4×10^{-16} S m⁻¹, 11 h value) more than two orders of magnitude lower than that of the reference LDPE (3.8×10^{-14} S m⁻¹, 11 h value), at an electric field of 32.5 kV mm⁻¹. For the other modified nanocomposites, which had an IPD larger than the theoretical value, the logarithmic value of the conductivity increased linearly with increasing IPD, indicating that the interparticle distance is the dominating factor in LDPE/ZnO nanocomposites controlling the conductivity-suppression level.

The surface modification of the nanoparticles may possibly also affect the charge mobility of the nanocomposite. The ZnO nanoparticles with a large specific surface area have surface defects and hydroxyl groups, suitable for charge-carrier trapping, but the surface coating may cover these defects and hydroxyl groups, and reduce the possibility of trapping charges/charge carriers. Among the silane coatings, only the C8 modification resulted in optimal surface modification, with a low degree of silsesquioxane coverage and high coating porosity,^[9,78] which should provide additional sites for charge carrier trapping in addition to the ZnO hydroxyl groups and defects. The ZnO nanoparticles produced by aqueous precipitation contain

residual counter-ions, which make the ZnO hydroxyl groups and defects inactive.^[24,25] The surface-modified and ultrapure ZnO nanoparticles without residual counter-ions^[9,26] showed a high conductivity-suppression (Figure 5c, ultrapure nanoparticle: STE-C8) in spite of their large IPD. Hence, the surface functionality not only controls the IPD but also probably the number of sites available for charge-carrier trapping. However, it is difficult to conclude that the presence of functionalities and counter-ions on nanoparticles surfaces is responsible for changing the charge mobility and DC conductivity, based on only DC conductivity data.

In general, for simple geometrical reasons, micrometer-sized particles form a smaller specific interfacial area with the polymer matrix than that formed in a nanocomposite system. However, it is possible to synthesize hierarchical micrometer-sized ZnO particles having a specific surface area (16.7 m² g⁻¹) comparable to that of 25 nm sized nanoparticles (33.6 m² g⁻¹). Figure 5d,e shows highly structured porous ZnO flower-shaped particles (≈1 µm) consisting of 20 nm thick nanosheet assemblies.^[24] These high surface-to-volume ratio particles have 99.9 vol% porosity with pores larger than 10 nm, into which the LDPE chains can fit. Hence, the large SSA of these microparticles make it possible to achieve an interfacial area as large as in the nanoparticle systems. The micrograph of the LDPE composite filled with hierarchical micrometer-sized ZnO particles shows, after acid etching, that the LDPE crystal lamellae are growing out of the particle surface at an angle normal to the particle surface (Figure 5f).^[9] This indicates a nucleation effect in the presence of ZnO particles during the crystallization of LDPE. The nucleation effect can be evaluated using DSC by recording the crystallization onset temperature under a low cooling rate (1 °C min⁻¹) (Figure 5g). The LDPE filled with hierarchical micrometer-sized ZnO particles showed an onset temperature higher than that of the nanocomposite system, which confirms that the porous micrometer-sized ZnO particles provide a particle/LDPE interfacial surface area similar to that of a nanoparticle/LDPE system.

Figure 5h shows the electrical insulation capacities of LDPE and of the LDPE composites with 3 wt% C8-modified ZnO nanoparticles or micrometer-sized ZnO particles. A significant decrease in conductivity was observed, not only for the nanocomposite but also for the microcomposite. It is suggested that the large interfaces in this particular microcomposite provide a trap density similar to that in the nanocomposite. The interparticle distance in the microcomposite (≈4 µm) is more than an order of magnitude greater than that in the nanocomposite system (≈140 nm). The interaction volume for charge trapping associated with these microparticles cannot cover this large interparticle distance. However, the changes in morphology of the polyethylene/spherulitic structure induced by the nucleation of the hierarchical micrometer-sized ZnO particles (providing smaller spherulites) can possibly extend further into the LDPE matrix. Smaller spherulites give a larger spherulite boundary area, which probably adds more traps for charge carriers. Hence, it is suggested that the reduction in charge mobility of hierarchical microcomposites is due to formation of charge traps from both large interfaces and spherulite boundaries. As stated earlier in Section 2.1, the hypothesis has not been confirmed due to the fact that the presence of

traps in the interspherulite boundaries is difficult to detect experimentally since it is difficult to see the isolated effect of spherulite size and structure, i.e., without having all other factors unchanged.

Although semiconductive ZnO particles, with a bandgap of 3.3 eV,^[24] have a potential for use in LDPE nanocomposites for ultrainsulating HVDC cables; there is an upper limit of possible nanoparticle loading, due to their nonlinear electrical behavior in high electric fields. This behavior is insignificant with metal-oxide nanoparticles such as MgO and Al₂O₃, with bandgaps of, respectively, ≈ 7.8 ^[86] and 7.6 eV,^[87] i.e., similar to that of LDPE. Hence, these can be used in larger amounts than is possible with semiconductive or conductive particles. **Figures 6a,b** shows the structure of LDPE nanocomposites filled with, respectively, 6 wt% of unmodified and C8-surface modified 60 nm MgO nanoparticles.^[12] The surface modification of MgO nanoparticles improved the degree of dispersion in the LDPE matrix, as in the case of modified ZnO particles. In the case of surface-modified MgO, the optimal dispersion also occurred with a relatively large amount of nanoparticles (9 wt%, structure not shown here). In order to gain more information about the degree of dispersion, 3D distributions of unmodified and C8-modified MgO nanoparticles in LDPE were modeled, based on dispersion statistics taken from microscopy images (Figures 6c,d). The unmodified and modified MgO nanoparticles showed IPDs of, respectively, ≈ 300 and 40 nm. The short distance in the latter case indicates perfect

dispersion of MgO nanoparticles, and is, in fact, close to the percolation value reported for LDPE/ZnO micro- and nanocomposites.^[29] The LDPE nanocomposites filled with 9 wt% of C8-surface modified MgO nanoparticles showed a DC conductivity more than one order of magnitude lower than that of pristine LDPE, in an electric field of 32.5 kV mm⁻¹ (Figure 6e). The use of a comparable amount of ZnO nanoparticles can, however, lead to electric breakdown of LDPE nanocomposites in high electric fields, which confirms that the dielectric characteristics of the metal-oxide nanoparticles themselves play an important role for the final insulating properties of the nanocomposites.

Figure 6e shows the DC conductivity of LDPE/MgO nanocomposites as a function of average interaction radius for different nanoparticle surface coatings and different loadings.^[12] Here, the interaction volumes are considered as imaginary spheres with a specific interaction radius around the MgO particle/agglomerate (Figures 6c,d). The first indication of a dramatic reduction in conductivity was observed at an interaction radius of 775 nm, based on 0.1 wt% of surface-modified MgO nanoparticles (Figure 6e). In the case of the LDPE nanocomposite with unmodified MgO nanoparticles, this large decrease occurred at a higher particle content (1 wt%), but the conductivity of the latter system was 10 times lower than that of the former system. Hence, it is suggested that the highly active surface area of unmodified nanoparticles has a more significant impact on the trapping of charge carriers and on the

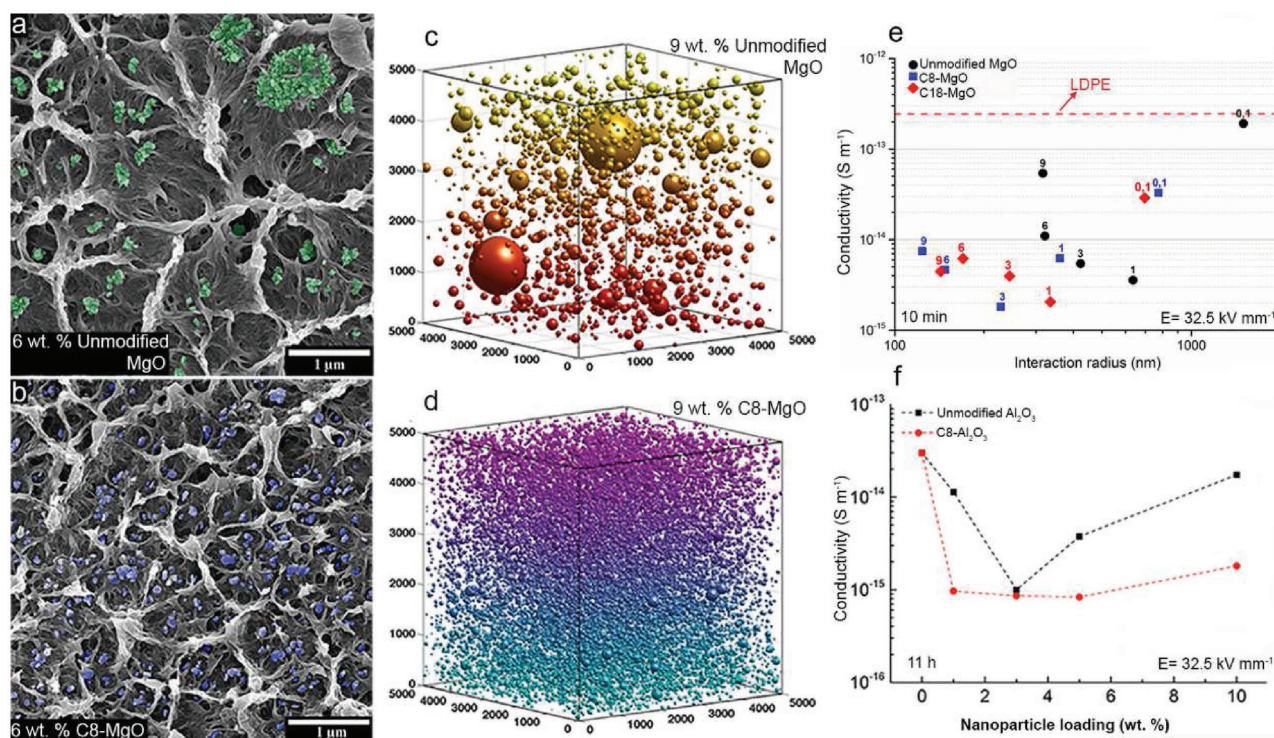


Figure 6. a,b) SEM image of LDPE nanocomposites with 6 wt% unmodified and C8 surface modified MgO nanoparticles. c,d) 3D size distribution model of 9 wt% unmodified and C8 surface modified MgO nanoparticles in LDPE nanocomposites. e) The DC conductivity of LDPE nanocomposites as a function of the interaction radius for the MgO nanoparticles with different surface functionality. a–e) Reproduced with permission.^[12] Copyright 2016, Royal Society of Chemistry. f) The DC conductivity plotted as a function of nanoparticle weight percent for LDPE nanocomposites based on unmodified and C8 surface modified 50 nm Al₂O₃ nanoparticles. Reproduced with permission.^[88] Copyright 2017, IEEE.

resulting reduction in conductivity, than short IPDs due to surface modification. However, the LDPE nanocomposites based on unmodified nanoparticles with a higher loading (6–9 wt%), showed a large increase in conductivity while the conductivity level was essentially unchanged when modified particles were used. This behavior was suggested to be related to the presence of adsorbed water in frequently appearing large agglomerates of unmodified MgO nanoparticles, which provided a conduction path through the agglomerates and a local increase in conductivity.^[38] The role of the nanoparticle surface activity and interparticle distance were essentially the same with Al₂O₃.^[88] Figure 6f shows the DC conductivity of LDPE nanocomposites based on unmodified and C8-surface-modified Al₂O₃ nanoparticles with an average size of 50 nm, plotted against the nanoparticle content under a constant applied electric field. The maximum reduction in conductivity occurred at 3 wt% for nanocomposites with unmodified Al₂O₃ nanoparticles and at 3–5 wt% with surface-modified nanoparticles. The unmodified nanoparticles provided a larger active surface area, whereas a shorter interparticle distance was obtained as a result of the surface modified nanoparticles. Hence, this indicates that the combined effect of surface activity and interparticle distance yield optimal conductivities at approximately the same particle content in these cases.

To summarize, the interface between the particles and the polyethylene matrix most likely play an important role for the resulting electrical properties. It is possible that polar groups on the particle surface and a “disturbed” polymer molecular order at the particle surface provide charge traps, which lower the charge mobility and the electric conductivity. The active particle surface-to-volume ratio should be as high as possible. This is achieved with very small particles (nanoparticles) or with large particles with a porous structure. Larger porous particles act as nucleation sites for crystallizing polyethylene, yielding smaller spherulites and a large spherulitic interfacial area. A hydrophobic surface treatment of the particles, improves their dispersion in the hydrophobic polymer, but reduces the active surface. An optimal result seems to be achieved by a “porous/incomplete” hydrophobic coating (see the C8 coating in Figure 4d). Although the actual choice of nanoparticle system (semiconductive or insulating) does not affect the insulation properties at a 3 wt% particle content, which is below the percolation threshold, the choice is important at higher loadings, and in high electric fields. Using insulating particles (MgO and Al₂O₃) it is possible to go to higher loadings and higher electric fields without electrical breakdown than when using semiconducting particles (ZnO). When choosing a particle system, the possibility of a particle-related additional moisture uptake in the nanocomposite must be considered.

3. Nanocomposites Properties

The short-term electrical performance of polyethylene nanocomposites based on metal-oxide nanoparticles has been shown to be better than that of the pristine polymer, but in order to implement these materials in real cables, it is important to know whether other properties, such as stability against

degradation and mechanical properties are affected by the presence of nanoparticles.

3.1. Interactions between Polar Species and Interfaces

Generally used extruded HVDC cables based on crosslinked LDPE (XLPE) exhibit better mechanical properties, i.e., higher stress cracking resistance and heat deformation properties, than pristine LDPE.^[8] Albeit, the scope of this article is mainly about LDPE nanocomposites, the effects of crosslinking byproducts needs to be considered at an earlier stage, before developing XLPE-based nanocomposites. The crosslinking of LDPE is initiated by an organic peroxide, typically dicumyl peroxide, which yields several low molar mass byproducts, e.g., methane, acetophenone, α -methylstyrene, and cumyl alcohol.^[8,89,90] If they are present in LDPE, these species may act as charge carriers, and increase the DC conductivity according to Equation (1).^[90] New traps are also possibly generated in their presence. These by-products are currently either removed by degassing the cables or avoided by using crosslinking agents that do not produce by-products. The long times and high temperatures necessary for degassing make the process less cost-effective and the complete removal of contaminants and ionic species is probably not feasible for the cable industry. Crosslinking agents, such as *n*-octane, which produce no byproducts, can be used in combination with irradiation;^[91] but the final crosslinking density is not sufficient to provide the required mechanical performance. The addition of nanoparticles with a high active surface area and a large number of hydroxyl groups make it possible to adsorb the detrimental polar species and other contaminants.

The adsorption of polar species (water, acetophenone as a byproduct from peroxide-vulcanization and Irganox 1076 as a phenolic antioxidant) on pristine and surface-modified metal-oxide particles is shown in Figure 7a,b, where the specific surface area and the content of accessible surface hydroxyl groups are factors that determines the adsorption capacity of metal-oxide nanoparticles. In fact, it is possible, by controlling the pore structure in the interface, to selectively adsorb small and detrimental molecules, i.e., water and crosslinking byproducts, on the metal-oxide nanoparticles surfaces while leaving larger beneficial molecules (antioxidants) intact in the polymer matrix. The surface-pores can be produced either during the synthesis^[10,25,26] or by surface functionalization by silanes after the synthesis of the metal-oxide nanoparticles.^[78] In order to utilize the adsorption properties of metal-oxide nanoparticles within the LDPE matrix, the diffusion kinetics of the polar species in the nanocomposites need to be considered. According to Fick's first law, the flux of small molecules in the LDPE nanocomposite is controlled by the diffusivity and the concentration gradient of the molecules in the LDPE matrix. The presence of nanoparticles possibly results in a constrained polymer layer at the interface, which reduces the overall diffusivity in the system. This may, however, be less important than the rapid “drag” of small molecules to the highly active particle surfaces.^[40,92]

In order to stabilize LDPE against oxidative degradation, the phenolic antioxidant, Irganox 1076, is commonly used in small

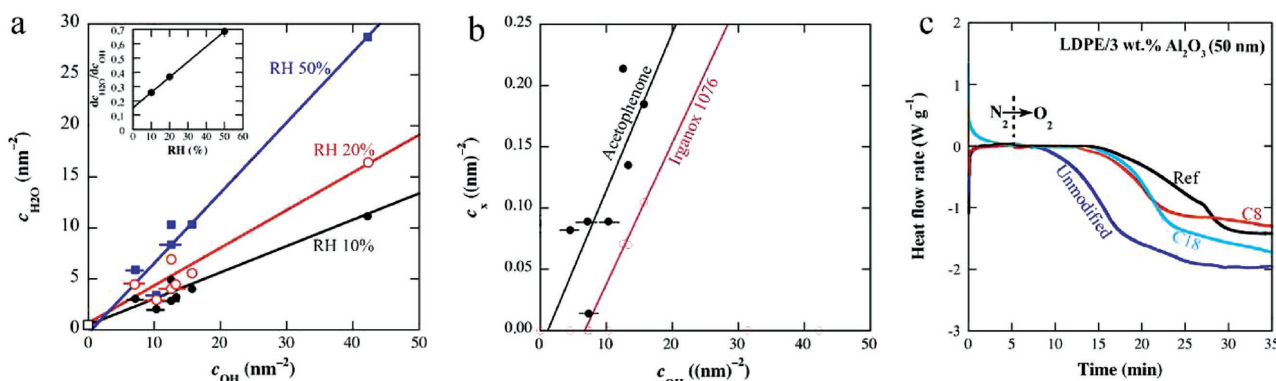


Figure 7. Adsorption of polar species, i.e., moisture, crosslinking by-product, and phenolic antioxidant onto metal-oxide nanoparticles. The coated nanoparticle samples are marked with a horizontal line. a) Moisture uptake as a function of hydroxyl group concentration for different metal-oxide nanoparticles (ZnO, Al₂O₃, and MgO) at 25 °C. The inset graph shows the slope of the fitted lines that are plotted as a function of RH. b) The concentration of adsorbed species (x) onto metal-oxide particles surfaces as a function of concentration of hydroxyl group concentration in a heptane medium; Irganox 1076 with a fitted line to only the three data points with nonzero values. c) DSC thermograms recorded at constant temperature (190 °C) for OIT measurements of nanocomposites that contain 2 ppm of Irganox 1076. Reproduced with permission.^[33] Copyright 2016, Elsevier.

quantities.^[33,93] The addition of antioxidants to LDPE leads only to a negligible change in the morphology and the electrical properties.^[94] On the other hand, the Irganox antioxidants containing hydroxyls and π -electron rich phenol groups, show a strong tendency to be adsorbed onto metal-oxide nanoparticle surfaces in nanocomposite systems, via hydrogen bonding and electrostatic interaction.^[33] This high degree of adsorption of antioxidants onto the nanoparticle surfaces, reduces their efficiency in protecting the polymer from oxidation during long-term service.^[95] The degree of adsorption of Irganox 1076 onto Al₂O₃ nanoparticles with different surface functionalities in an LDPE matrix was assessed by determining the oxidation-induction-time (OIT) at 190 °C in a DSC (Figure 7c). The LDPE with uncoated Al₂O₃ nanoparticles showed a much shorter OIT, but the nanocomposites which contained coated nanoparticles (C8- and C18-) showed a smaller reduction in OIT. If the nanoparticles have a rough surface with surface pores smaller than are needed to host the large antioxidant molecules, however, their adsorption on the particle is expected to be small, while water and small polar species can still be adsorbed. Tailoring the surface functionality of the metal-oxide nanoparticles is thus an important way to control the adsorption on the particles; in the best of cases, trapping ionic/polar species/impurities while leaving the antioxidants in the bulk material to prevent oxidation of the polymer.

3.2. Mechanical Properties and Cavitation: Effects of Interfacial Strength and Particle Dispersion

In polymer matrices, nanoparticles can have either positive or negative effects on the mechanical properties. Debonding between the polymer matrix and the particles leads to cavitation and poor mechanical properties (Figure 8a),^[32] and can also lead to the development of electric trees and a lower electric breakdown strength, but the use of surface coatings on the particles improve the matrix/particle interfacial bonding. Liu et al.^[32] surface treated Al₂O₃ nanoparticles with silanes having one (C1), 8 (C8), and 18 (C18) carbons attached to the silicon

atom before adding the nanoparticles to the polyethylene matrix. The position of the second yield point in the tensile stress-strain curve indicated the strain where the onset of cavitation occurred (Figure 8b), which is contemporary with the geometric instability in the sample referred to as “necking”. It was shown that the onset of cavitation/necking in LDPE/Al₂O₃ nanocomposites occurs later when the particle surface is modified with the C18-terminated silane than with the C8-silane, although they show a similar degree of dispersion (Figure 8b).^[32] The polymer-particle adhesion was found to decrease in the order: C18, C8, C1, and uncoated. In the LDPE/ZnO nanocomposites, on the other hand, the best dispersion was obtained with the C1- and C8-terminated silanes, whereas the C8 and C18 silanes showed the highest strain for the onset of cavitation, which was better than the uncoated and C1-coated nanoparticles.^[9] The C8- and C18-silanes probably did not cover the whole particle surface and possibly allowed for polymer molecules to adhere to the particle surface between the silane molecules and partially entangle with the coating molecules. A good dispersion combined with a “strong” interface seem to give the best mechanical properties.^[9,10,32,96]

The fractal number, d_s , of the nanocomposite systems, determined by simultaneous small angle X-ray scattering (SAXS) and tensile testing, can yield more information on the interfacial structure.^[96] Below 50% strain, the highest interface fractal number is found in the LDPE/Al₂O₃ nanocomposite filled with uncoated nanoparticles; the nanocomposites based on coated nanoparticles show smaller values and decreased in the order: C18, C8, C1. In the case of uncoated nanoparticles, which are nonwetting to polyethylene, only a part of the nanoparticle surface area adheres to the polyethylene matrix, which leads to large fractal number at the interface. The fractal number for all nanocomposites decrease until $\approx 50\%$ strain and then level out (Figure 8b, inset), indicating that the particle surface went from a “rough” to a smoother surface, indicative of debonding between the particle and the polymer matrix or of a filling of the particle–polymer porous interface. The polymer/particle interfaces were directly observed on a nanoscale using X-ray ptychographic tomography of strained

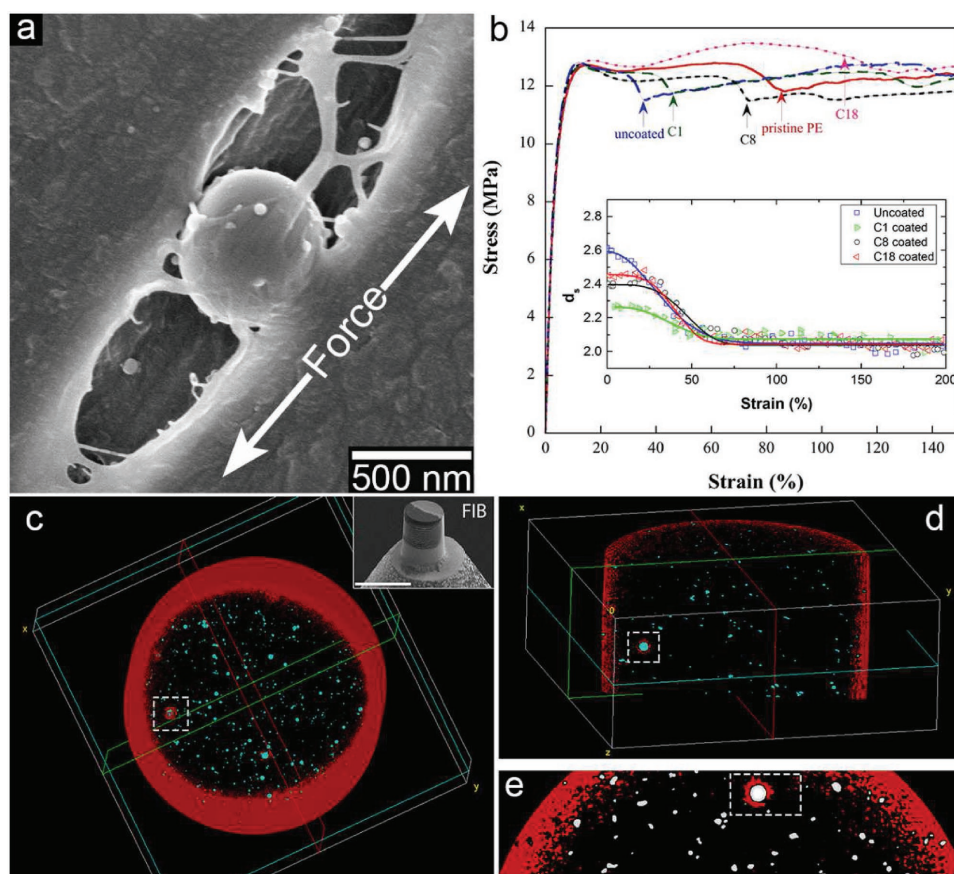


Figure 8. a) SEM image of a LDPE nanocomposite based on Al_2O_3 nanoparticles stretched to 100% average strain. Reproduced with permission.^[32] Copyright 2015, Elsevier. b) The stress–strain curve (Reproduced with permission.^[32] Copyright 2015, Elsevier.) and corresponding interface fractal dimension values, i.e., d_s (Reproduced with permission.^[96] Copyright 2017, Elsevier.), plotted as a function of strain for an LDPE nanocomposite containing 3 wt% Al_2O_3 with different functionalities. c) 3D rendering of the transitional region (between un-necked and necked area) in a strained LDPE nanocomposite containing 3 wt% of C8- Al_2O_3 nanoparticles using X-ray ptychographic tomography. The bright and red regions (not the red wall) correspond, respectively, to the nanoparticles and low-density regions of the LDPE (cavities). The inset shows the examined pillar of the nanocomposite created by focused-ion-beam-SEM (scale bar is 30 μm). The box with dashed lines shows a nanoparticle surrounded by the strain-induced low-density regions of the polyethylene. d) Section through the pillar at the highlighted nanoparticle in the y–z plane. e) Section through the pillar at the highlighted nanoparticle in the x–y plane. c–e) Reproduced with permission.^[96] Copyright 2017, Elsevier.

LDPE/ Al_2O_3 nanocomposites (Figures 8c–e).^[96] Ptychographic nanotomography is a coherent diffraction imaging technique that can show cavity features in polyethylene with a nanometer resolution.^[96,97] In this technique, the phase images are used to achieve a 3D map of the electron density distribution in the specimen, which in turn can be transformed into a mass density distribution.^[97] This technique cannot be used in situ, similar to the combined tensile/SAXS technique. Hence, the sample is examined after the tensile test, and it is not possible to see the real time evolution of the mass density during the cavitation process. There is therefore a risk that some of the cavities around the particles are closed after relieving the tensile stress and not observed by the tomography method. The strain-induced low-density regions, shown by red color, are found mainly around big nanoparticles indicating their large cavitation. Poor dispersion (large agglomerates) and large particles lead to the formation of larger cavities (and more brittle materials) under tension. The degree of dispersion examined

at the nanoscale has shown its effect on the mechanical properties at the macroscale.

4. Charge Transport Mechanisms in Nanocomposite-Based Insulation Materials

4.1. Current Models and Mechanisms of Electric Behavior in Insulating Nanocomposites/Nanodielectrics

4.1.1. Bipolar Charge-Transport Model

Although often combined, we present the bipolar charge model, the induced dipole moment model and the multicore model under separate headings here. The bipolar charge transport model has been used to describe the charge transport in LDPE and its nanocomposites under an applied electric field (Figure 9a).^[98,99] The behavior of the charges can be described

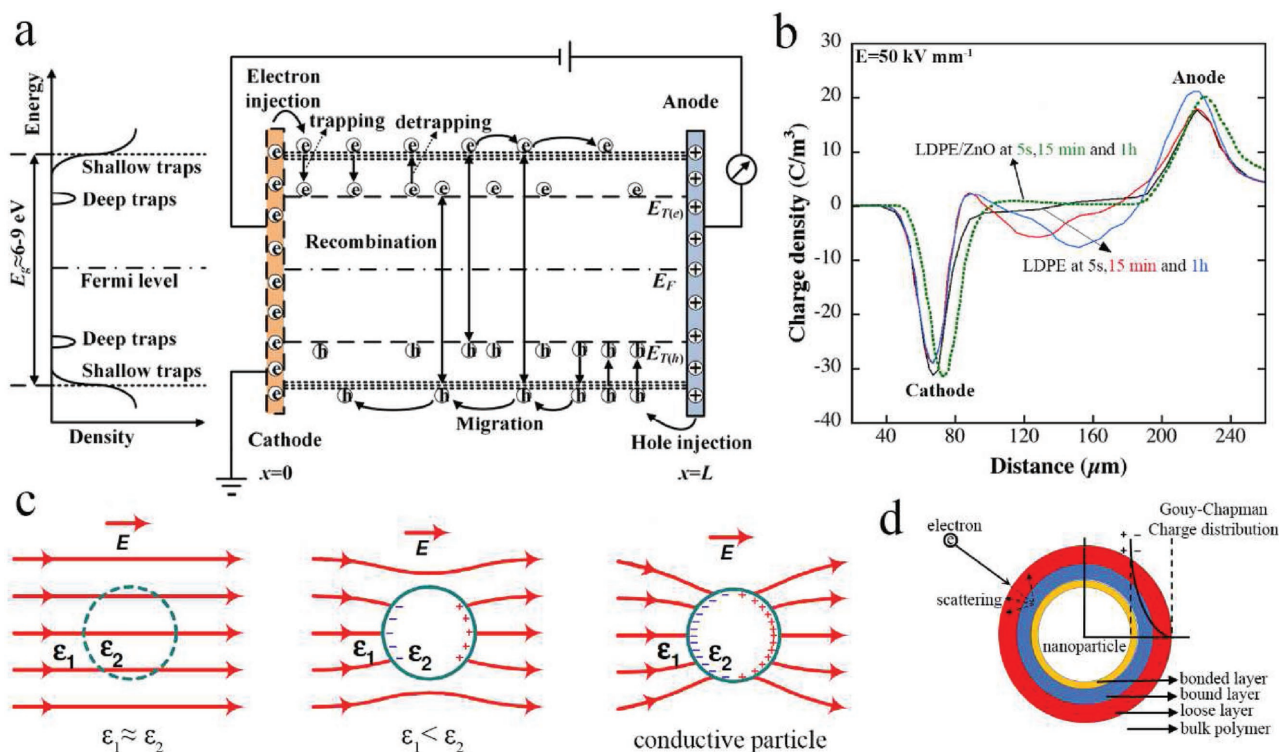


Figure 9. a) Illustration of the bipolar charge injection and transport in nanodielectrics under an applied electric field. Reproduced with permission.^[98] Copyright 2015, IEEE. b) Space-charge distribution at different times for LDPE and its nanocomposite filled with 7 wt% ZnO nanoparticles; data replotted from ref. [102]. c) the mutual induction model showing induced dipole electric charge and line of electric force distributions around a spherical filler with different dielectric permittivity from polymer matrix. Redrawn from ref. [37]. d) The multicore model for polymer nanocomposites. Redrawn from ref. [35].

by a set of self-consistent relationships, involving charge advection-reaction, charge transport and spatial time-dependent potentials (Poisson's equation). Under an applied electric field, electrons are injected at the cathode and move through the insulating material toward the anode. At a low electric field, Schottky thermionic emission governs the charge injection, where the charge injection can be described by the use of an injection barrier that depends on temperature, the contact potential barrier, the electric field and the dielectric permittivity.^[98] In high electric fields, charge injection occurs through field-emission tunneling. There are both shallow (localized states) and deep traps in which the electron can be trapped. The time in the trap, i.e., between trapping and de-trapping, increases with increasing trap depth (larger negative energy). At the anode, holes are injected into the insulation and move toward the cathode, and hole traps with different depths also exist. Besides trapping, electrons and holes may recombine in the dielectric, an event which occurs more frequently at higher temperatures, as observed by simulations of charge transport in dielectrics.^[100] The migration of charges depends not only on the depth of traps and on the degree of recombination but also on the trap density. The probability of trapping in deep traps increases with increasing charge carrier mobility (the set input value in the modeling, not the effective mobility) and decreases with decreasing permittivity. The probability of detrapping decreases exponentially with the trap-energy (depth), and the rate of recombination increases with increasing electron and

hole mobility. The first order charge dynamics thus depends on the charge trapping, detrapping, and recombination rates.

It is considered, as mentioned earlier, that charge traps form at structural/physical defects in materials (e.g., branches and kinks in polymer chains), as well as at chemical defects (e.g., double bonds and polar “defects” such as carbonyl groups (keto, ester, and carboxyl groups)).^[37] For pristine polyethylene, calculations have shown that the residence time increases with increasing trap depth and lower electric field.^[101] It has been estimated that both the trap depth and the trap density are larger for nanocomposites than for the pristine polymer. The trap depth is estimated to be in the order of 1–5 eV for a LDPE/MgO nanocomposite at a high electric field, which is higher than the calculated trap depths of chemical defects such as carbonyls.^[37] The density of traps has also been suggested to increase when nanoparticles are incorporated (e.g., traps due to polar chemical groups on the particle surface), which reduces the space charge accumulation in nanocomposites.

An example of the difference in charge accumulation in LDPE with and without nanoparticles (ZnO) is shown in Figure 9b, as monitored by the pulsed electro-acoustic (PEA) method.^[102] Both electrons and holes are immediately injected from the cathode and anode separately and symmetrically within 5 s after the voltage is applied. After 15 min in pristine LDPE, a large charge accumulation is detected close to the cathode, which later on drifts toward the anode. In the LDPE/ZnO nanocomposite, however, there was no charge

accumulation during 1 h, which indicates a significant space charge suppression in this material. Based on the bipolar charge transport model, Min et al.^[98,103] considered that the presence of nanoparticles in LDPE lowered the space charge accumulation and the DC conductivity primarily through a change in the injection barrier, whereas the change in trap density and depth primarily affected the dependence of the conductivity on the nanoparticle content. The increase in injection barrier gave a lower space homocharge mobility because the presence of nanoparticles in the vicinity of the cathode leads to a lower local “maximum electric field” and consequently a higher apparent breakdown strength than in the pristine polymer. On the other hand, the nanoparticle-induced deepening of traps is not sufficient to explain the fact that the space charge accumulation in the nanodielectrics is lower than in the pristine polymers.^[37] Hoang et al.^[100] also used the bipolar charge transport model to model the charge transport in nanodielectrics, and they showed that the reduced charge current in the nanocomposite was due partly to an increase in the injection barrier at the electrode–insulator interface. The reduced charge current in the nanocomposite was also claimed to be due to a reduction in the effective charge mobility and an increase in trap depth and density. They also showed that the presence of nanoparticles led to a lower space charge accumulation and local field enhancement. In the bipolar model, several assumptions are made which are not necessarily correct. Among the input parameters, e.g., the injection barrier and deep trap energy and density (in ref. [98]) and/or detrapping energy barrier and deep trap density (in ref. [100]), are considered being the same for electrons and holes. The nature of the “macroscopic” interface between two samples of different or the same material also has a strong influence on the accumulation of space charges and overall charge carrier mobility. Hu et al.^[104] showed that negative charges accumulate at the interface between two “identical” polyethylene plates after the removal of an applied DC field. A heterogeneous material may also experience Maxwell–Wagner polarization due to location-dependent permittivity and electric conductivity.^[104]

Homocharge accumulation (observed by PEA) close to the cathode in unfilled LDPE increases the local field close to the cathode at short times, but Lau et al.^[105,106] showed that heterocharges accumulate near the cathode at longer times. Heterocharges are considered to originate mainly from the ionization of impurities, e.g., water and crosslinking byproducts (XLPE).^[102] Hence, not only holes and electrons move in an electric field. Besides the electric field, any concentration gradient of the ion itself affects the chemical potential and the driving force for the “net” diffusion of the ion. The kinetics of diffusion are dependent on parameters such as the size of the diffusant and the molecular mobility of the polymer environment.^[107] Fuqiang et al.^[80,108] showed that the electrical properties of epoxy below and above the glass transition temperature (T_g) depended primarily on electron transport and electron accumulation below T_g , but on ion transport and ion accumulation above T_g . The positive ions accumulated rapidly close to the cathode, since neutralization via mass (ion) transport across the electrode/insulator contact was essentially absent. As heterocharges accumulate near the cathode, the local electric field near the cathode is enhanced resulting in a potential

barrier reduction for electron injection.^[102] The local electric field is thus enhanced by homocharges at short times and heterocharges at long times, which results in a larger charge mobility in a pristine polymer. When SiO₂ nanoparticles and Al₂O₃ microparticles were added to the epoxy matrix, a lower space charge accumulation, complex permittivity, and conduction current were observed above the glass transition temperature, which was ascribed to steric hindrance by the particles, although it was not clear whether this was a geometric impedance or a constraining effect on the molecular mobility of the polymer.^[80] The adsorption of ions onto the particle is another plausible explanation if nanoparticles are added into pure polymers, they are also likely to adsorb heterocharges.^[102,109] Whereas Tanaka^[110] attributed the reduced space charge accumulation in nanocomposites to the introduction of shallow trapping states, Tian et al.^[102] showed, using a discharge current technique and PEA, that there was a significant increase in the amount of deep traps (≈ 0.8 – 1 eV) and a decrease in charge mobility when ZnO nanoparticles were introduced into the LDPE. When electrons/holes are injected from the electrodes, they are considered to be trapped in trapping states at the insulator/electrode interface, where charges are too close to the electrode to be resolved by current space charge detection methods. This shows that the space charge distribution has a complex behavior with time, and more information from direct observations on a nanoscale are required in order to clarify the charge transport mechanisms.

Even though the bipolar model has been successful in logically showing, e.g., the increase in breakdown field/strength with increasing trap density and depth, the model, because of its simplicity, has clear limitations in the description of the dielectric systems and therefore also its prediction power.^[99] For example, it is assumed that for each system there is only a single (no distribution) trap density and depth (for both traps in the particle/polymer interphase and within the polymer matrix) and any release or transfer of energy associated with charge injection/extraction is not considered. In addition, ion transport is not included and, important for the breakdown strength, the impact ionization and heterocharge accumulation is not considered.

4.1.2. Induced Dipole Moment Model and the Role of Permanent Dipoles

When an electric field is applied across a dielectric material, permanent dipoles as well as induced molecular dipoles reorientate, which results in a change in the local electric field. Permanent dipoles from chemical defects such as carbonyls contribute with only a small dipole moment,^[37] but these dipoles may possibly act as charge trapping sites when the electric field is high. Because of their low dipole moment, the traps are however calculated to be shallow (typically ≈ 0.3 eV) and act over a relatively short distance (≈ 1 nm). Dipoles induced by nanoparticles, on the other hand, lead to large dipole moments and therefore possibly deep traps because of the large distance between the particle parts of opposite charge (Figure 9c). The trap depth and the size of the induced dipoles associated with the nanoparticles is expected to depend on their permittivity,

size, and shape. In the induced dipole-moment model it is considered that these deep traps at the interface between the polymer and the nanoparticle lead to an overall space charge reduction. Takada et al.^[37] considered the space charge reduction to be associated with the trap depth, and found that the reduction increased with increasing particle size (up to 200 nm) and increasing particle relative permittivity. This implies that a conductive particle with a high relative permittivity should be more effective than an insulator or semiconductor particle, in reducing space charge accumulation. Both size and permittivity effects suggested by this model seem to be controversial; macroscopic experimental evaluations of different nanocomposite systems show different results. For example, the insulating properties of LDPE/MgO nanocomposites calculated from macroscopic measurements (conductivity, permittivity, dielectric loss factor, and space charge distribution), are superior to those of microcomposites.^[111] This has also been shown for LDPE/ZnO nano- and microcomposites.^[9,29,30,109] Roy et al.^[112] showed that XLPE composites with micrometer-sized SiO₂ had a lower breakdown strength than those with nanosized SiO₂. It seems that the formation of more trap sites on the nanofiller surface (larger SSA) leads to better insulating properties, but this has not been considered completely in the Takada model. However, it was concluded theoretically that the particle shape is important: a sphere yielding deeper traps than an ellipsoidal particle.^[37]

It is evident in the induced dipolar moment model that a conducting particle becomes a stronger dipole along an applied electric field than an insulating/semiconducting particle. A reduction in DC conductivity of approximately two orders of magnitude was observed for LDPE when 3 wt% MgO, Al₂O₃ (insulating) or ZnO (semiconductive) particles was added (see Figures 5c and 6e,f), which suggests that the nanoparticle/polymer interface is the place hosting deep charge traps.^[9,12,28] However, as mentioned earlier with contents greater than 6 wt% an electrical breakdown occurs in the system with semiconducting ZnO nanoparticles, but not in the systems with the insulating MgO or Al₂O₃. DFT calculations on an LDPE/MgO system suggest that the interior particle structure is also important, at least to some extent.^[113] There are evidently several mechanisms that can be responsible for the reduction in conductance of an insulating matrix like polyethylene when it is doped with nanoparticles.

4.1.3. Multicore Model

Tanaka et al.^[35] proposed a multilayer description of the particle–polymer interface (Figure 9d). It consists of a “bonded” layer closest to the particle, a “bound” layer further out and finally a “loose” layer. With a model using these three layers, it was possible to describe the breakdown strength in polyamide; and it was claimed that the stronger bonds between the particle and the polymer in the bonded and bound layers yielded a higher breakdown strength.^[35] Tanaka et al.^[35] also suggested that a diffuse Gouey–Chapman layer occurs over all three layers; a phenomenon observed in “electrolytes” which involves a gradual decrease in the electrostatic potential away from a charged particle. Li et al.^[114] proposed a similar three-layer

interface model to explain the increase in deep trap energy/content leading to a lower charge carrier mobility, the suppression of space charges and an increase in breakdown strength in different nanocomposite systems. The Li's model consists of, from the particle and outward, a bonded, a transitional and a normal region. The molecules in the bonded region are connected to the particle with strong bonds, such as covalent, ionic or hydrogen bonds. In the transitional layer, the polymer molecular mobility is altered yielding a change in, e.g., the glass-transition temperature, crystallinity, cohesive energy, and content of shallow traps. Being simple and intuitive, it has to be remembered that the multicore models are just models and the presence of different discrete layers is currently not easy to verify experimentally. Even though EFM methods have to a certain extent this capability, they have their own limitations.

The interfacial regions have also been claimed to be scattering centers for charge carriers with the ability to reduce the energy of the mobile charges, which result in an overall reduced charge accumulation (Figure 9d).^[37,114,115] The large improvement in breakdown strength of XLPE/silica over XLPE was suggested to be due to the scatter of charge carriers at the very large specific surfaces of the nanoparticles, which limits space charge accumulation.^[112] In addition, it was proposed that the interface between untreated silica and LDPE becomes quasi-conductive (as indicated by dielectric spectroscopy data), which likely enables a redistribution of charges to limit the local electric field and increase the breakdown strength.^[112] Yao et al.^[116] ascribed the improved electric insulation properties in the presence of SiO₂ particles in polyethylene to a similar phenomenon, i.e., a field-smoothing effect (less accumulation of space charges) inside the nanocomposite when the external field is changed.

Roy et al.^[112] investigated the effect of surface treatment of silica with polar and nonpolar functionalities on the electrical properties of XLPE/silica nanocomposites. It was suggested that the relatively small improvement in, e.g., voltage endurance with the addition of a particle surface coating was due to a lower content of defects at the particle surface (due to the chemical bonding between the coating and the particle) and/or to the introduction of traps (due to polar end-groups (–NH₂)). An important factor is also that the “interface” seems to extend when a surface treatment is used.^[117] As already mentioned in Sections 2 and 3, a more porous coating on metal-oxide nanoparticles likely leads to that more polar groups (hydroxyl) remain on the surface and that charge carriers are able to reach these “inside” the coating. Bell et al.^[118] reacted π -conjugated electroactive surface ligands (e.g., anthracene) onto silanized silica nanoparticles using poly(glycidyl methacrylate) brushes before mixing the particles with an epoxy matrix. They reported a higher dielectric breakdown strength than when only uncoated particles were used. They claimed that this was due to improved dispersion (long brushes) and a higher electron trap density and depth (π -conjugated parts).

Calculations on LDPE/MgO nanocomposite systems using the induced dipole model showed that the trap well around a 50 nm MgO particle can be as wide as 50 nm (using 0.1 eV as the cut-off) under a very high electric field (100 kV mm^{−1}).^[37] The interface/interphase has a polymer structure which is perturbed by the presence of the nanoparticle, which may result

in, e.g., a lower chain mobility and chain entanglement density.^[117] Therefore, it was argued that the size of the interfacial region should be of the order of the polymer radius of gyration (10–20 nm). Tanaka et al.^[35] suggested in their multilayer model, that 20–50 nm large particles should have an interface thickness of 10–30 nm. The second “bound” layer in their model was considered to be on the order of 2–9 nm thick. Kubyshkina et al.^[113] estimated, based on DFT calculations, that the interface “effects” between polyethylene and MgO particles are able to extend at the most 40 nm into the bulk of the polyethylene from the MgO surface. It must be remembered that the size of the simulated system is limited. Smaller interfacial dimensions were observed for LDPE/SiO₂ nanocomposites when the molecular structure was investigated using SAXS.^[116] The SAXS-determined interface increased from 2.8 to 3.5 nm when the temperature was raised from 30 to 100 °C and it was claimed that this was due to a partial melting of the polymer.^[116] Hence, different interfacial thicknesses are

predicted by simulations, theoretical models, and material characterization. There are probably several reasons for this, which include differences in: (i) the external conditions (e.g., the size of the electric field), (ii) the actual system (particle, particle surface treatment and polymer matrix), and (iii) the way the interface properties has been calculated/measured.

4.2. Electronic Structure Simulations of Interfaces in Polyethylene Nanocomposites

Kubyshkina et al.^[113,119] investigated theoretically the effects of nanoparticles (MgO) with different surface structures on the electrical behavior of polyethylene. **Figure 10a** (left and right) shows polyethylene interfaces with three different MgO crystal planes (atomic termination), i.e., MgO (100), hydroxylated MgO (111), and silicon-treated MgO (111), together with their projected densities of states. Here, the polyethylene was modeled

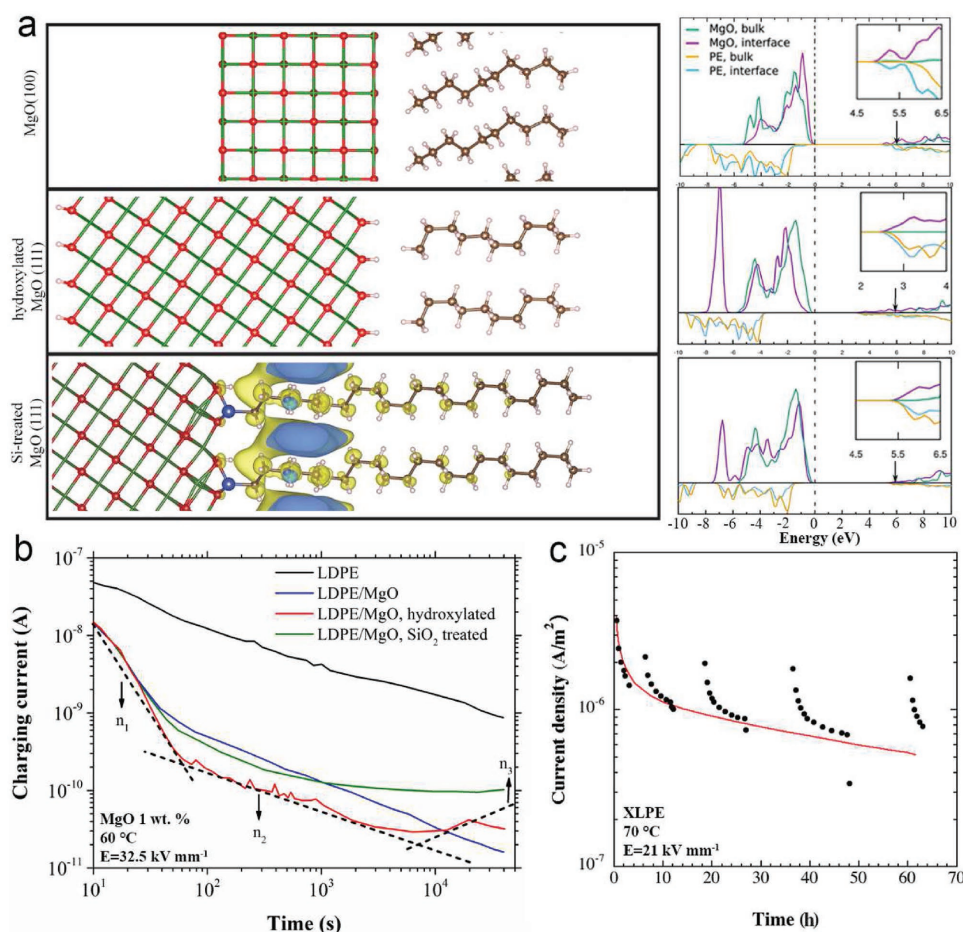


Figure 10. Electronic structure modeling of different LDPE/MgO interfaces together with experimental DC conductivities. a) geometries representing the modeled systems (left), together with their projected densities of states (right); states in 0.2 eV interval from the bottom of the conduction band are visualized for the extended Si-treated MgO (111)/polyethylene interface, as yellow isosurfaces with cyan sections; magnesium, oxygen, hydrogen, carbon, and silicon atoms are, respectively, shown in green, red, white, brown, and blue colors. The zero energy in the projected density of states corresponds to the valence band edge, with respect to the vacuum level, of bulk MgO (6.6 eV). The inset figures show the magnified conduction band edges and the arrows points to the bottom of the conduction band of bulk MgO. a) Top right part: Reproduced with permission.^[119] Copyright 2017, AIP Publishing; remainder of image: Reproduced with permission.^[113] Copyright 2017, AIP Publishing. b) Charging current as a function of time, for polyethylene samples with and without MgO with different crystal planes. Reproduced with permission.^[10] Copyright 2016, American Chemical Society. c) Current density versus time for an XLPE sample with a constant (red line) and switched (filled circles) voltage; data replotted from ref. [120].

using $C_{10}H_{22}$ chains. It was concluded that the presence of the particle/polymer interface does not introduce any new states/traps due to changes in the polymer conformational structure. It was also suggested that a double layer (positively charged particles and a negatively charged surrounding shell) may form at the interface, due to changes in the band structure. From DFT calculations it was shown that for all the systems studied, ((100) and (111) surfaces with hydroxyl-groups and (111) partially surface-coated with the silicon), the position of the conduction band of the MgO surface/interface was aligned with the polyethylene conduction band position (Figure 10a, right). It was suggested that this originated from the similarity in the conduction states in the two materials. The modeling also showed that in polyethylene, the conduction states are located in vacuum between the polymer chains and overlap with surface states of the conduction band of MgO in the vacuum outside the MgO crystal (Figure 10a, left). The bandgap of the nonpolar MgO (100) surface, was smaller than the value for the bulk MgO, and shifted to larger negative energies. A shift to larger negative valence and conduction band-edge values was also observed in polyethylene, but the size of the bandgap was the same as in the bulk. In the presence of the more polar hydroxyl-terminated (111) MgO face, the shift in the conduction band energy to larger negative values was greater than that of the valence band shift leading to an even smaller bandgap for the MgO surface/interface than for the MgO (100) case. Again, the bandgap of polyethylene in the interface was the same as in the bulk. However, the band bending observed in polyethylene at the interface is suggested to mean that additional electrons move from the bulk to the particle hydroxyl-rich surface where they are trapped in deep traps. It is therefore suggested that the interaction between polyethylene and the MgO crystal surface leads to formation of new traps in the bulk polymer.

DFT calculations also show that the electronic bandgap structure and energy of states changes significantly for a polyethylene/MgO nanocomposite system when a silicon atom with a dodecane tail (polyethylene equivalent) is attached covalently to a hydroxyl-functionalized MgO (111) crystal surface (Figure 10a, left).^[113] The calculated bandgaps of the MgO surface and its interface with LDPE, as stated above, were lower than that of the MgO crystal core, but increased to the crystal core bandgap value when coated particles were considered. The coating replaces hydroxyl-related states in the conduction band and, since the surface is interrupted by silicon atoms, the electron mobility along the particle surface is reduced. Although simulations/calculations as given above makes it possible to study/estimate, e.g., the size of the bandgap and distribution of electronic states in great detail, care must be taken when evaluating the results since the size of the system studied, for computational reasons, has to be small and the structure idealized (refer to Section 2.1).

4.3. Long-Term Conduction Current in Polyethylene Nanocomposite Insulation Materials

Although the PEA measurements on LDPE nanocomposites can give informative results about the charge distribution on a macroscale, it is evident from many studies cited that it is

important to monitor the charge behavior over much longer times than is normally done. Figure 10b shows the charging current of LDPE/MgO composites and of an unfilled LDPE. The surface functionalities of MgO are controlled through synthesis and the heat treatment of $Mg(OH)_2$,^[10] and the surface functionalities of synthesized MgO nanoparticles are the most resembling structure of the simulated atomic termination in the last section, which can be found in the literature. The charging current for the composite systems was always lower than that for the pristine LDPE throughout the measurement period (<11 h). Among the nanocomposite systems studied, LDPE filled with hydroxylated MgO showed the largest charge suppression and the lowest DC conductivity. The hydroxylated MgO had the highest specific surface area for charge trapping, but the polymer/nanoparticle interfacial areas were probably similar for all the nanocomposite systems, which means that the charge suppression due to trapping of charges should be similar for all nanocomposites, if the nanoparticles had the same intrinsic characteristics (traps, bandgap, and dielectric properties). However, it appears that the MgO phase without hydroxyl groups, obtained by heat treatment at 1000 °C, had a higher conductivity due possibly to crystal lattice imperfections resulting from the thermal annealing. On the other hand, the simulations showed that the calculated bandgap for the hydroxylated MgO surface was the lowest. The experimental results also showed that the hydroxylated surface of MgO was more prone to transform into a hydroxide phase than the nonhydroxylated surfaces.^[10] This suggests that the hydroxylated MgO can be considered to have a pure MgO core with several layers of $Mg(OH)_2$ at the surfaces. It is probable that these experimental input-data would yield simulation data that are more realistic.

The DC conductivity (charging current) of all the nanocomposite systems decreased with time following a power-law behavior, $I(t) \propto t^{-n_i}$ (Figure 10b). The power factors (n_i , $i = 1, 2$, and 3 as indicated in Figure 10b) were obtained from the slope of the fitted line of $\log I$ versus $\log t$. The unfilled LDPE showed a linear conductivity reduction (on log scale), whereas the composite systems showed nonlinear relationships. Lau et al.^[105] related this behavior to changes in the charge transport mechanism (three stages or slopes were detected and shown by n_1 , n_2 , and n_3 in Figure 10b). The results were interpreted as follows; a low power factor indicates the formation of trapping sites for space charge carriers in the semicrystalline matrix, whereas a high power factor is consistent with a reduction in charge carrier mobility (tunneling and hopping) at the nanoparticle/polymer interface.^[105] In the first stage, the trap-limited transport (both Ohmic and space-charge-limited-current (SCLC)) involves a gradual filling of traps. As the traps are filled, the filled traps become increasingly important leading mainly to trap-limited SCLC charge transport (stage 2). When all the deep traps are filled the conduction increases with time (stage 3). This increase in current is accompanied by a replacement of homocharges by heterocharges at the cathode, as mentioned previously. This shows that it is important to assess the nanocomposite insulation properties over long periods of time. It is, of course, necessary to avoid stage 3 behavior in cable insulation materials, which are intended to remain in service for several years. It is suggested that the greater slope in stage 1 for the nanocomposite than for the filler-free polymer is due to

charge trapping processes in sites in the polymer–particle interfacial region.^[105,106]

As mentioned above, the LDPE shows only a linear decay of charging current with time (on log scale within the 11 h experiment, Figure 10b). A similar slope has also been observed for crosslinked polyethylene (XLPE).^[120] In order to see the influence of the charging current history, the voltage (applied electric field) was switched on and off during a 60 h measurement. After switching on the voltage, the polarization current decayed within 2–3 h to a “master” curve (Figure 10c). This important finding suggests that the charging current decay is a voltage-independent process. Ghorbani et al.^[120] also presented, in relative terms, long charging-current experiments (up to 150 h) and they observed both stage 2 and 3 for XLPE and pristine LDPE (1.2 mm thick sheets, 70 °C). The behavior of the two polymers was similar over short periods of time, and the slopes of the charging (leakage) current versus time curves for XLPE and LDPE were similar (a slope of ≈ -0.5 on a $\log(\text{current})$ – $\log(\text{time})$ axes, before 50 h). After longer periods of time, after stages 2 and 3, a broad shoulder was observed followed by, in the case of XLPE, a new decrease in current. For LDPE, the experiment was ended while still in stage 3 (increase in current versus time). It is not possible to say whether the XLPE feature with a broad shoulder will appear for the LDPE system at times beyond the time-frame of the experiment. Recent results have shown, however, that the shoulder also occurs for pure LDPE (measured over 12 d), and even for polypropylene; but these results were obtained in a low-field region (7 – 10 kV mm^{−1}).^[121] It thus appears that stages 2 and 3, and the associated mechanisms are not unique to particle-containing systems. Further, intermittent and continuous applied voltages have been shown to yield an overlapping current–time curve (master curve), which suggests that the change in charging current with time is a thermally activated physical relaxation process in the material rather than a purely electrical feature.^[121] The charging current decreases with time even in the absence of an electric field. At 60–70 °C, LDPE partially melts and this changes its properties and affects the physical relaxation processes (in the region of the α -relaxation).^[122] Further support for an electric-field-independent process has been obtained by aging samples for different times (up to 10 months) at 50 °C.^[121] It was observed that the charging current (at 30 kV mm^{−1}), measured over 24 h, decreased with increasing total aging time. At high electrical fields (30–60 kV mm^{−1}) the shoulder observed in the $\log$ (charging current) versus $\log$ (t) diagram was absent or much smaller than in the low electric-field experiments. In addition to time-dependent relaxation processes in the material, the semicrystalline morphology has been shown to affect the electrical properties, as also discussed above using TSC in Section 2.1.

The TSC measurements on LDPE/silica nanocomposites indicated that the α -relaxation, appearing above T_g in polyethylene, was affected by the presence of nanosilica.^[117] In the presence of the particles, the intensity of the relaxation increased and shifted to a lower temperature, and it was argued to be due to a greater amount of charge traps in the particle/polymer interface of the nanocomposite than in the pure polymer, which lacked this interface. Consequently, during poling,^[117,121] it was suggested that more charge carriers are caught in the

polymer/particle interface traps, which subsequently, during heating, means that more charge carriers enter the conduction band yielding a more extensive thermal-induced relaxation. The dielectrically determined α -relaxation is considered to be due to longitudinal chain transport through the crystal,^[123] but as indicated here by the TSC data, also seems to be affected by the electrical properties of the interface. Yao et al.,^[116] using dynamic mechanical tests, observed an increase in the α -relaxation temperature when SiO₂ nanoparticles were added to LDPE, and argued that this was because of a greater crystal thickness in that system. It must be noted here though that the polyethylene α -relaxation, determined by dielectric measurements, does not necessarily capture the same molecular motions as with dynamic-mechanical tests (where often two α -relaxations are observed).^[122] This may explain the differences in the observed results by Smith et al.^[117] and Yao et al.^[116] regarding the shift of the position of the α -relaxation temperature with the addition of nanoparticles. Li et al.^[114] showed that the peak intensity of the α -relaxation (with TSC) passed through a maximum when the Al₂O₃ content was increased from 0 to 10 wt%. Hence, if the amount of trapped charges is scaling to the intensity of the α -peak, the amount of trapped charges, for whatever reason, seems to occur at an intermediate particle content.

4.4. Direct Observation of Charge Transport in Interfaces on a Nanoscale

There are only a few reports on the direct observation of the “electric” interface. Borgani et al.^[11] used ImEFM to study the surface potential of C8-Al₂O₃ insulating nanoparticles dispersed in a 100 nm thick LDPE film, as shown in Figure 11a–c (the method was shown earlier in Figure 1b). Without a DC bias (on top of the AC voltage that makes the probe vibrate) the less electronegative particles are positive and the surrounding polymer matrix is negative. The experiment clearly showed that the interaction zone (with the interface) was greater than the diameter of the particle and that its potential decreased gradually toward the matrix. For a 50 nm Al₂O₃ particle, the interface thickness was of the order of 30 nm. Using a positive DC bias (up to 60 kV mm^{−1}), the interfacial and particle potentials increased and the size of the interface increased (more holes are injected from the ImEFM probe to the particle/interface region). In the figure, the interfaces of the two particles overlap. When a negative voltage is applied, hole extraction from the vicinity of the particle occurs. This is a slower process than the hole injection and means that the particles become invisible, having the same potential as the matrix. In the presence of the nanoparticle, localized energy states are suggested to become available in the energy gap close to the conduction band, referred to as shallow traps.^[11] The presence of shallow traps due to the presence of nanoparticles was discussed earlier based on DFT simulation results.^[113,119] This makes it favorable to inject holes from the ImEFM probe, which leads to a positively charged particle. With a positive DC bias, the Fermi energy decreases at the probe which allows even more injected holes and a higher surface potential at the particle. With a negative DC bias, holes are extracted from the traps by the probe until all traps are emptied. The extraction can be slower than

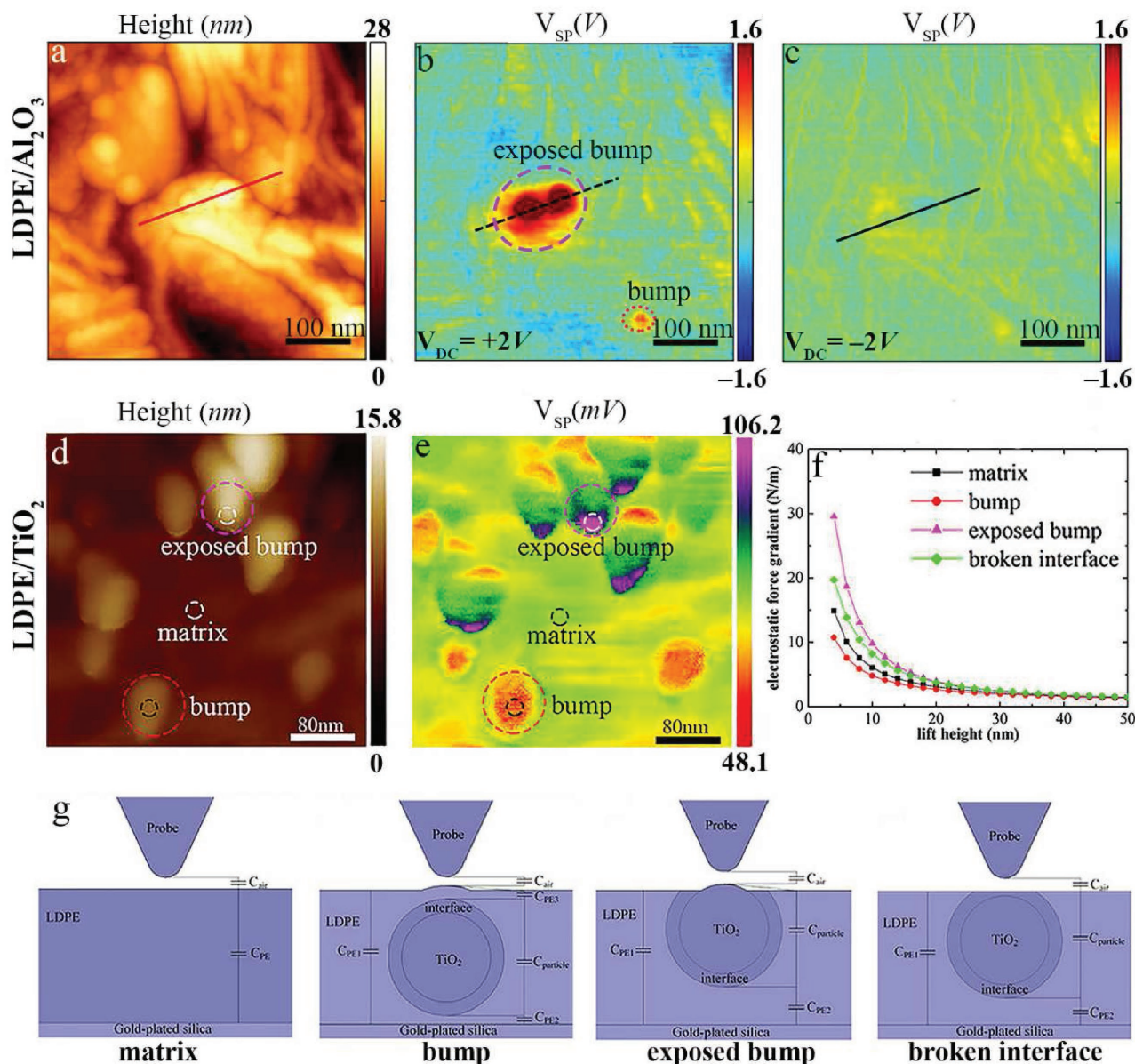


Figure 11. a) AFM height image of an LDPE/ Al_2O_3 nanocomposite. b,c) ImEFM surface potential V_{SP} under a positive (b) and negative (c) 2 V DC bias. Reproduced with permission.^[11] Copyright 2016, ACS. d,e) AFM height image (d) and ImEFM surface potential (e) of an LDPE/ TiO_2 nanocomposite. f,g) Simulated electrostatic force gradient plotted as a function of probe lift height for different particle/matrix structures (f) and illustration of the ImEFM measurement on different particle/matrix structures (g). Reproduced under the terms of the CC-BY 4.0 licence.^[75] Copyright 2016, The Authors. Published by Nature Publishing Group.

the injection because of the small difference between the energy of the traps and the Fermi energy at the probe. Thus, a time-dependent change in the sample surface charge occurs with a negative bias, as was observed by consecutive scans by the probe on the sample surface (not shown).

Peng et al.^[75] used ImEFM, with a DC bias over the AC, to study the interface in LDPE nanocomposites based on semi-conductive and relatively high dielectric permittivity TiO_2 nanoparticles (Figure 11d,e). From their ImEFM data, it is possible to estimate the size of the interface, and the interaction zone seems to be larger than the particles similar to the LDPE/ Al_2O_3 case above. They investigated the dielectric behavior of the

LDPE matrix with nanoparticles embedded in different ways (Figure 11g). Contrary to the LDPE/ Al_2O_3 nanocomposites, the LDPE "bump" over the TiO_2 nanoparticles showed lower surface potential than the LDPE matrix. The bump was suggested to be the bound polymer layer in the multicore model. From their EFM data it was claimed that it was also possible to determine indirectly a relative permittivity of this type of interface, which was reported to be lower than that of the matrix, explained by a lower flexibility of the polymer chain leading to less polymer dipoles orienting along the field (slow dipole polarization). On the other hand, the exposed surface of the TiO_2 nanoparticles showed a greater surface potential than that of the LDPE matrix

due to the high permittivity of the pure nanoparticles (fast ionic polarization). Peng et al.^[75] also used the finite element method to calculate the interaction force (electrostatic force) for the LDPE matrix and embedded nanoparticles (Figure 11f,g). They found that the experimental and simulation results were consistent if the interface is considered as a part in the multicore model. The specific size of the interface was estimated from the experimental results. This type of interface is modeled as a “broken interface”; the interface has been partially removed while the nanoparticles have not been completely exposed. The simulation in Figure 11f reveals that the electrostatic force gradient of the “broken interface” is between that of the “bump” and “exposed bump.” This suggests that this interface, with the thickness comparable to the size of particles, has intermediate dielectric properties for LDPE/TiO₂ nanocomposites. Hence, it is essential to determine the interface structure (size and dielectric properties) on a nanoscale, since this most likely plays an important role for the electrical properties of nanocomposite insulation materials. It should be remembered that the EFM methods described in this chapter are surface methods and do not necessarily reflect/capture a 3D case, i.e., inside the bulk, or other surface-side of particles in order to examine the induced dipole by nanoparticles with different dielectric properties (see Figure 9c).

5. Conclusion and Future Outlook

The nanodielectrics based on polyethylene and metal-oxide nanoparticles are receiving tremendous attention and research interest for use in HVDC insulation materials, which must in the future withstand high level of voltage and mechanical stresses. To fulfill this application of polyethylene/metal-oxide nanocomposites, discreet “bottom-up” strategies are required to obtain strong interfaces between metal-oxide nanoparticles and the polyethylene matrix. It has been indicated that the “strength” of the interfaces is one of the main factors that determine the final long-term mechanical and electrical properties of these insulating materials. Considerable achievements have been made in recent years in improving the interfaces in polymer nanodielectrics, but less attention has been paid to the chemical and dielectric properties of the pure components. In this progress report, we have shown how bulk and surface properties of functionalized metal-oxide nanoparticles (i.e., size, crystal lattice, crystal atomic termination, surface hydroxyl group concentration, high-purity, and surface coating porosity), are likely to influence the interface structure and the final nanocomposite properties. It is of paramount importance to monitor experimentally the nano-/microstructures of pure components and interface, in addition to simulating the charge transport in nanocomposites. The identification of charge transport mechanisms from the molecular to the macroscale helps in designing new materials with superior long-term properties.

It is shown in this progress report that the use of pure functionalized metal-oxide nanoparticles and highly porous micrometer-sized metal-oxide particles improve the electrical insulation properties of polyethylene, using charging current measurements over a relatively limited time period. In many cases the charging current never reached a steady-state value

within the experimental time frame. Most of the charging current–time curves are characterized by a rapid initial decrease in current, due possibly to polarization effects, followed by a slower decrease. In the presence of particles, however, the whole curve is, in general, lower (meaning lower conductivity) than that of the particle-free polymer. However, in both cases, toward the end of the experiment, the current may increase with an eventual decrease again, showing a shoulder. The mechanisms of these obscure effects need to be clarified, especially since the expected service life of a cable is many years. It is of course of utmost importance that the improved insulation properties achieved with the added particles are refined over the entire service lifetime. Hence insulation measurements over long periods of time are needed to reveal the long-term effects of the particles, as well as of the polyethylene insulation itself. The effects of moisture also need to be evaluated. Even though the current cables are sealed with a metal outer barrier layer it cannot be assumed that the material is “completely” moisture-free. Even though polyethylene is hydrophobic, and takes up only a very small amount of moisture, this amount could still be sufficiently high to affect the insulation efficiency. With the use of metal-oxide particles, the moisture uptake increases, and this effect has to be considered in future experiments.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

charge transport, high-voltage insulation, interfaces, metal-oxide nanoparticles, polyethylene, surface coatings

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